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模擬機器人自動化倉儲系統提升履單效率之動態訂單分配與儲位選擇策略研究

**Dynamic Order Assignment and Pod Selection
Strategies to Enhance Fulfillment Efficiency in Robotic
Mobile Fulfillment Systems: A Simulation Approach**

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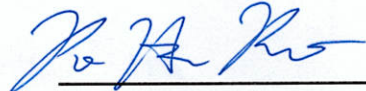
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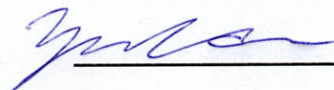
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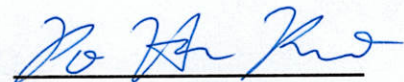
 



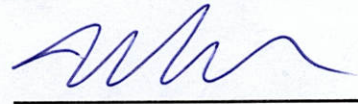
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ABSTRACT

This study presents an improved decision framework for enhancing order picking efficiency within Robotic Mobile Fulfillment Systems (RMFS) by addressing two critical operational processes: Pick Order Assignment (POA) and Pick Pod Selection (PPS). While previous research has typically addressed these decisions with limited coordination, such static approaches fail to account for the dynamic nature of warehouse operations, often leading to inefficiencies, including suboptimal order placement for the picking process, reduced pile-on rates, and longer order completion times.

To address these challenges, this study proposes Future-Aware order assignment strategy that, by incorporating awareness of order completion status and the next available bin, evaluates SKU fulfillment rates across all picking stations based on the pods that are already assigned, queued, or in transit. This mechanism enables the system to assign orders dynamically, based on both current available bin and near-future bin availability, which supports better SKU fulfillment and contributes to faster order completion times.

For pod selection, the system employs a scoring method based on the total SKU demand at each picking station, prioritizing pods that can fulfill the highest number of order lines coverage for current orders at the picking station. This approach shifts the focus from individual orders to overall picking station demand, aiming to maintain high pile-on performance and support more efficient order fulfillment.

The effectiveness of the proposed framework is validated through simulation experiments implemented in the NetLogo environment. Results demonstrate that the proposed approach achieves approximately 18.80% improvement in throughput, a 11.50% reduction in order completion time, and 34.86% increase in pile-on efficiency to a conventional baseline. The approach adapts to real-time warehouse conditions, demonstrating strong applicability in resource-constrained environments and offering a practical solution for enhancing operational efficiency and order fulfillment performance.

Keywords: RMFS, Pick Order Assignment, Pick Pod Selection, Future-Aware Strategy, Dynamic System

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CHAPTER 1

INTRODUCTION

1.1 Background

The rapid expansion of e-commerce, driven by transformative global events such as the COVID-19 pandemic and high-volume shopping festivals like China's Double 11, has significantly intensified the operational pressures on warehouse and logistics systems. As of 2025, e-commerce accounts for 21% of global retail sales and is projected to reach 22.6% by 2027 [1]. The number of online shoppers reached 2.77 billion in 2025 about 33% of the global population, and is expected to grow to 2.86 billion by 2026. This surge is supported by a global e-commerce infrastructure exceeding 28 million stores, with annual sales projected to surpass \$6.8 trillion [1]. Consumer behavior is also shifting: 52% of online shoppers seek products internationally, 34% shop online weekly, and 99 % rely on reviews during purchasing decisions. This shift toward frequent, small-sized, and highly heterogeneous online orders has highlighted the limitations of traditional warehouse models and furthered the shift toward automation-driven systems.

Traditional picker-to-parts systems require human workers to walk extensive distances through warehouse aisles to retrieve items [2], [3]. Although straightforward to implement, such systems are notoriously inefficient: up to 70% of a picker's time is spent walking rather than retrieving products, and order picking alone accounts for approximately 55% of total warehousing costs, making it the most labor-intensive activity in warehouse operations, as shown in Figure 1.1 [4]. In modern e-commerce contexts, characterized by a large variety of SKUs, small order lines, and fast delivery expectations, these inefficiencies become significant operational bottlenecks [5]. As businesses face rising labor costs, labor shortages, and higher throughput demands, reliance on purely manual picking methods is increasingly unsustainable.

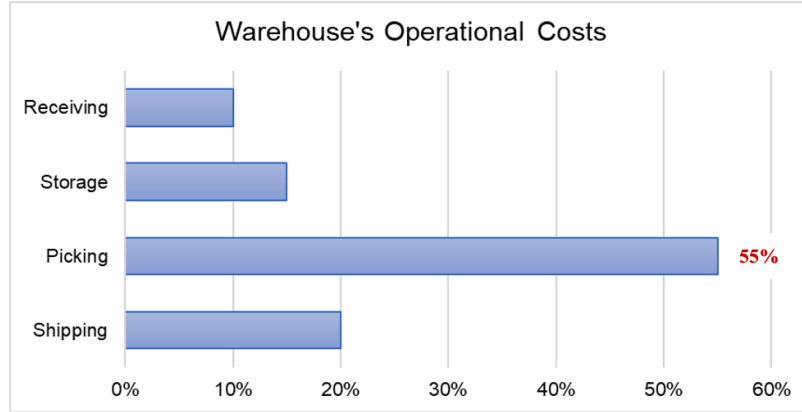


Figure 1.1 Warehouse operational cost by activity type [4]

To address these challenges, many fulfillment centers are transitioning from picker-to-parts to parts-to-picker systems, where autonomous systems deliver goods directly to stationary pickers. This shift improves picking productivity by reducing unnecessary travel, maximizing picker utilization, and better managing order complexity [6]. As a result, scalable, automated solutions have become essential for competitive e-commerce operations. The Robotic Mobile Fulfillment System (RMFS) is one such solution, an automated parts-to-picker approach in which autonomous robots transport movable storage racks, known as pods, between storage areas and picking or replenishment stations [7]. RMFS employs Automated Guided Vehicles (AGVs) to deliver pods containing required SKUs to picking stations for order fulfillment, return them to storage after use, or transfer them to replenishment stations for restocking inventory.

A notable milestone in the development of RMFS was Amazon's acquisition of Kiva Systems for 775 million USD in 2012, which transformed its distribution centers and revolutionized robotic warehouse operations [8]. Figure 1.2 illustrates a typical RMFS layout adapted from the Kiva model [9]. In addition, real-world deployments include Swisslog's CarryPick™, GreyOrange's Butler™, Alibaba's Cainiao™, and JD.com's Dilang system. Empirical studies have shown that RMFS can double or even triple picking productivity compared to conventional warehouse operations and improve order fulfillment speed by up to 50% over traditional warehouses [10].

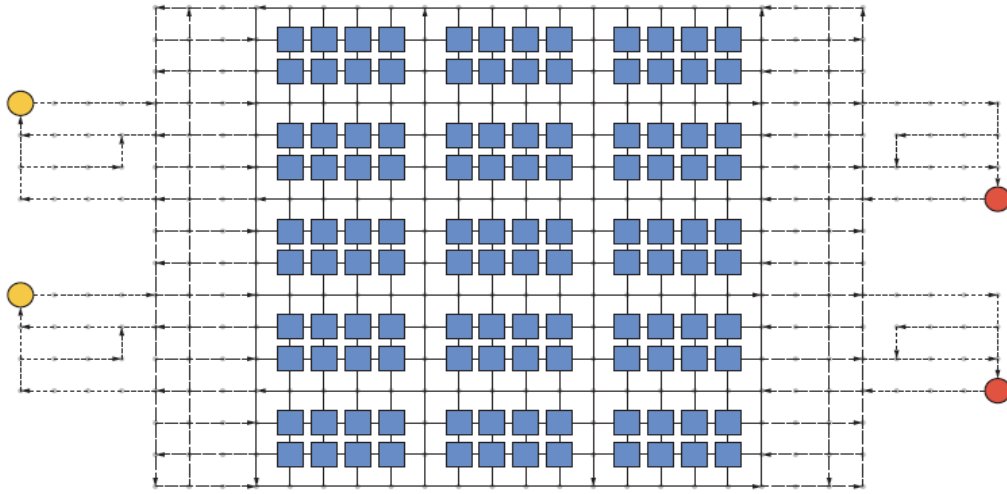


Figure 1.2 A simplified layout of the RMFS [9]

As companies scale up their operations, the adoption and advancement of parts-to-picker systems have accelerated, largely driven by increasing difficulties in sourcing sufficient skilled labor [11]. This shift is further motivated by three key operational pressures: limited labor availability, rising work volume and throughput demands, and the need to adapt to evolving business requirements [11].

The growing complexity of Robotic Mobile Fulfillment System (RMFS) operations introduces new decision-making challenges. Although many of these challenges have been addressed in prior studies, several key areas remain open for further improvement as the number of orders, pods, and robots continues to increase [12].

One of the core operational challenges lies in the Pick Order Assignment (POA) process, which determines how orders are assigned to available picking stations. Conventional or static POA methods often fail to account for real-time SKU availability and bin utilization, resulting in low pile-on rates, unnecessary pod movements, and increased picker idle time. Similarly, the Pick Pod Selection (PPS) process, which identifies which pods should be retrieved to fulfill assigned orders, critically affects pod utilization and overall system throughput. Inadequate pod selection can lead to suboptimal picking efficiency, longer robot travel distances, and system congestion, ultimately lowering throughput.

RMFS also requires coordination across multiple downstream processes. Once pods are selected, specific Autonomous Mobile Robots (AMRs) must be dispatched to transport the pod to the designated picking stations, this is known as Task Allocation (TA). To ensure smooth robot movement, Path Planning (PP) must be performed to avoid congestion and deadlocks. After the required SKUs are retrieved by the picker, the system determines appropriate storage locations for the pods return within the warehouse, a process referred to as Pod Repositioning (PR). Each of these stages adds complexity and necessitates real-time, system-wide coordination for optimal performance.

Prior approaches often relied on fixed priority rules, which may perform adequately under static conditions but are insufficient when dealing with dynamic inventory states, bin availability constraints, and real time system changes. As order volumes, SKU diversity, and system complexity increase, conventional static in RMFS face growing limitations in maintaining high throughput and efficient resource utilization due to the lack of coordination across decision-making components. An optimal solution at an individual decision layer does not necessarily lead to a globally optimal outcome for the entire system. Furthermore, as noted by Wurman et al. [13], all activities within a Kiva based RMFS occur in real time, without predefined time windows for offline optimization of operational level decisions. This real time responsiveness, requiring algorithms to continuously adapt to the current state of the system and resource availability. RMFS environments operate under highly dynamic conditions, where key system variables such as SKU inventory levels, unfulfilled order, robot availability, and pod locations change continuously. These fluctuations demand robust and adaptive decision-making strategies capable of responding effectively to evolving warehouse states.

To address these challenges, this study proposes a simulation-based, Future-Aware framework that supports dynamic order assignment based on real-time inventory conditions. The framework near-future bin availability across multiple picking stations to enhance order-to-station assignment decisions. For pod selection, a Station Match strategy is employed, utilizing picking station SKU demand profile to identify pods with high supply potential based on inventory level. By operating under dynamic system states, the proposed framework offers a robust and adaptable solution suitable for modern RMFS environments characterized by high order volumes and diverse SKU profiles. By continuously aligning decision rules with real-time system conditions, the approach aims to maximize throughput, reduce order completion time, and improve pile-on efficiency.

1.2 Objective

Modern e-commerce fulfillment centers face increasing pressure to process a high volume of small, diverse customer orders within increasingly stringent time constraints. To address these challenges, this research aims to develop and evaluate a simulation-based, fulfillment-driven decision framework for Robotic Mobile Fulfillment Systems (RMFS) that dynamically coordinates two critical operational processes: Pick Order Assignment (POA) and Pick Pod Selection (PPS).

This study adopts a dynamic order assignment mechanism that allocates orders to available picking stations based on both current and near-future system states. The assignment process considers SKU fulfillment rates, which are calculated by evaluating the inventory levels of active pods, defined as pods that have already been assigned, are queued, or are currently in transit to a station. By effectively capturing the interdependencies among orders, pods, and picking stations, the proposed framework surpasses traditional one-to-one assignment methods, enabling more responsive and efficient decision-making in dynamic operational environments. This forward-looking approach aims to identify the most suitable picking station by evaluating both current and projected bin availability. In parallel, the pod selection strategy evaluates candidate pods based on their ability to satisfy the aggregated SKU requirements of multiple orders at a given picking station, aiming to maximize the number of order lines fulfilled per pod visit.

This research aims to contribute a practical and scalable methodology that enables real-time decision-making for RMFS and provides warehouse operators with an adaptable solution for improving order-picking performance in modern e-commerce fulfillment environments.

1.3 Scope and Limitation

This section outlines the scope and limitation of the current research to establish a clear context for the study's methodology, simulation assumptions, and applicability:

- The simulation is implemented using NetLogo for agent-based modeling, Python for optimization algorithms, and Microsoft Excel for data tracking and analysis.
- The simulation utilizes data modified from real-world warehouse scenarios to ensure contextual relevance while maintaining model adaptability.
- The simulation assumes a sufficient volume of orders to ensure continuous picking operations
- The layout and configuration of the warehouse is fixed.
- Each order in the simulation contains multiple Stock Keeping Units (SKUs) with specified quantities, reflecting typical e-commerce order characteristics.
- Each bin at a picking station can accommodate only one order at a time.
- Each pod contains multiple storage slots, but each slot carries one type of SKU.
- A single SKU may be stored in multiple pods.
- A pod is limited to visiting a single picking station per trip. Multiple stops or hopping between picking stations is not allowed to maintain operational simplicity.
- All orders are assumed to be fulfilled within the same day, and no priority or urgent orders are considered in the simulation.
- Initial order mechanism is included to preload the system with orders that were placed outside of business hours or carried over from earlier.

1.4 Organization of Thesis

The current study is divided into five chapters, each covering a different component of the study. The organization of the study is as follows:

- Chapter 1 – Introduction

This chapter outlines the motivation, background, and research objectives. Additionally, provide the scope, assumptions, and limitations of the work.

- Chapter 2 – Literature Review

This chapter presents a comprehensive review of relevant literature in the field of Robotic Mobile Fulfillment Systems (RMFS), with focus on order assignment, pod selection, and related decision strategies.

- Chapter 3 – Methodology

This chapter presents the methodological framework used in the study, including the simulation design, system architecture, and underlying modeling assumptions. The performance metrics and experimental setup used to evaluate system effectiveness are also described.

- Chapter 4 – Results and Discussion

This chapter presents simulation results and key findings. Sensitivity analyses are included to evaluate the robustness of the proposed framework.

- Chapter 5 – Conclusion

This final chapter summarizes the contributions and findings of the study. It reflects on implications of the results and proposes directions for future research.

CHAPTER 2

LITERATURE REVIEW

2.1 Robotic Mobile Fulfillment System

The Robotic Mobile Fulfillment System (RMFS) is an advanced automated material handling solution designed to improve the efficiency and speed of order fulfillment, particularly in e-commerce environments. RMFS uses autonomous mobile robots to transport movable storage units, known as "pods", directly to human or robotic picking stations, replacing the traditional picker to goods approach. This transition enables faster item retrieval, better space utilization, and a significant reduction in human walking time across the warehouse. As shown in Figure 2.1, Kiva robots, one of the earliest RMFS implementations, demonstrate how mobile robots navigate warehouse environments to efficiently deliver pods to picking stations [14].

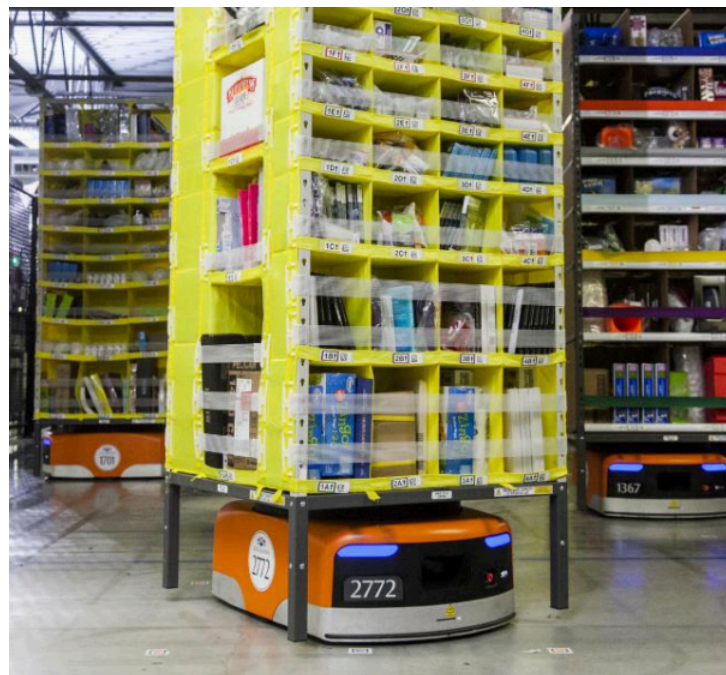


Figure 2.1 Kiva robots at the Amazon Fulfillment Center [14]

An RMFS typically includes physical infrastructure such as mobile robots, storage pods, and workstations. It also incorporates autonomous control software that manages robot movement, collision avoidance, recharging, path planning, and speed regulation. In addition, a warehouse management system (WMS) coordinates inventory flow, task assignment, order tracking, and synchronization between all system components [15].

Recent studies have consistently identified Robotic Mobile Fulfillment Systems (RMFS) as a key technology in the domain of warehouse automation. Enright and Wurman [16] provide a comprehensive overview of the optimization challenges involved in coordinating autonomous mobile robots within fulfillment environments, addressing sub-problems such as order assignment, replenishment, and robot allocation. Building upon this, Merschformann et al. [12] investigate the impact of operational decisions on fulfillment performance, emphasizing the importance of pick sequencing and storage location assignments. Their findings highlight the essential role of strategic decision-making in enhancing picking efficiency in RMFS operations. Furthermore, Boysen et al. [5] present a detailed survey of warehousing systems developed to meet the unique demands of e-commerce retailers, while Barros and Nascimento [17] review recent innovations, persistent challenges, and future opportunities in the evolving field of goods-to-person RMFS.

The storage and retrieval processes of a Robotic Mobile Fulfillment System (RMFS) are depicted in Figure 2.2 highlighting the flow of pods between storage areas and workstations. In the replenishment process, robots transport pods requiring restocking to the replenishment station, where SKUs are added based on received replenishment orders. Once restocked, the pods are transported back to the inventory storage area. When a picking task is issued, the system selects an appropriate pod and assigns a robot to perform a retrieval trip from the inventory. The pod is then delivered to the picking station, where the picker retrieves the required SKUs and places them into order bins according to order specifications. Following the picking process, the pod may be routed back to storage or redirected to another workstation, depending on system needs. This closed-loop process ensures coordinated storage and retrieval operations that support continuous and efficient material flow in the RMFS.

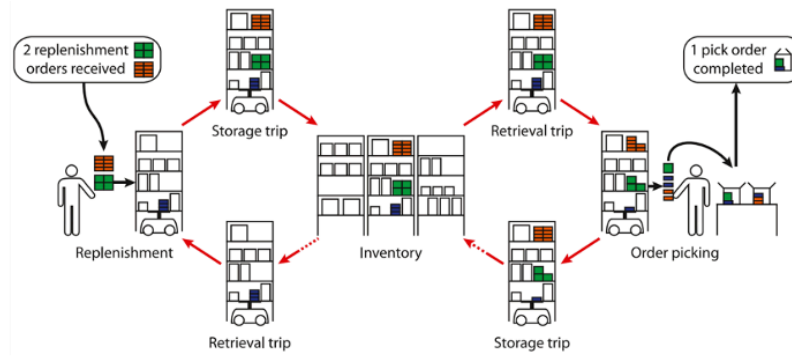


Figure 2.2 Operational cycle of a robotic mobile fulfillment system [12]

2.2 Decision Problem in RMFS

Decision-making in RMFS follows a hierarchical structure, encompassing strategic, tactical, and operational levels, each addressing different planning scopes and influencing system performance [12]. Table 2.1 outlines representative operational-level decision areas and their respective scopes.

Table 2.1 RMFS decision-making hierarchy and key problems

Strategic level Long-term design decisions during facility planning		
Key Problem	Problem Description	References
Workstation deployment	Deciding the optimal number and positioning of picking and replenishment stations to maximize process efficiency	[18]
Storage area configuration	Determining the spatial layout and sizing of the storage area to accommodate operational requirements	[10]
Tactical level Periodic setup choices at the start of a planning horizon		
Key Problem	Problem Description	References
AMR Robot	Deciding the number of AMRs required to handle expected workload and avoid bottlenecks	[19]
Battery scheduling management	Developing charging and usage schedules to prevent robot unavailability due to low battery levels	[20], [21]
Storage assignment	Assigning SKUs to pods based on demand or correlation, aiming to balance space usage and retrieval frequency	[21], [22], [23], [24], [25]

Operational level Real-time execution and control		
Key Problem	Problem Description	References
POA	Selecting which order from the backlog to assign to available bins based on readiness and station capacity	[7], [12], [15], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35]
PPS	Selecting the most suitable pod to dispatch to a picking station based on warehouse inventory	
ROA	Determining which replenishment station should handle the next incoming inventory restocking request	[12], [23], [36], [37]
RPS	Choosing an appropriate pod for replenishment items to the selected replenishment station	
TA	Assigning picking or replenishment tasks to available robots based on workload balance or travel efficiency	[38], [39], [40], [41], [42], [43], [44], [45]
PR	Also referred to as Pod Storage Assignment (PSA); deciding the optimal storage location for returning pods	[46], [47], [48], [49], [50]
PP	Generating efficient navigation paths for robots to avoid collisions and minimize travel time	[39], [51], [52], [53], [54]

2.3 Pick Order Assignment

Pick Order Assignment (POA) plays a pivotal role in Robotic Mobile Fulfillment Systems (RMFS) as it governs how orders are assigned for picking stations. Effective POA strategies directly influence throughput, pile-on performance, robot utilization, and overall system responsiveness.

Early contributions, such as those by Merschformann et al. [12] and da Costa Barros and Nascimento [17] provide comprehensive reviews that clarify how these decisions interact with other RMFS tasks. Boysen et al. [27], introduced joint optimization models for Pick Order Assignment (POA) and Pick Pod Selection (PPS), using Mixed-Integer Programming (MIP) focus on batching and scheduling of orders assigned to picking stations, as well as the ordering of pod deliveries, with the objective of minimizing the total number of pod-to-station visits required to fulfill the assigned orders. Valle and Beasley [55] formulate the problem of order assignment and pod (rack) selection in Robotic Mobile Fulfillment Systems (RMFS) as a mixed-integer linear program (MILP).

Their model simultaneously determines which orders are assigned to each picker and which mobile racks (pods) should be retrieved to fulfill those orders. Additionally, the authors address the rack sequencing problem, which determines the optimal order in which pods are presented to pickers, considering constraints such as bin capacity and the possibility of allowing pod revisits.

Yang and Rossomando [24] investigate the joint impact of storage assignment strategies and order batching policies in RMFS, introducing an integer linear clustering model that simultaneously groups items and batches orders based on similarity metrics. Their model aims to minimize robot movement, which directly correlates with order-picking efficiency in pod-based systems. Justkowiak and Pesch [33] develop a column-generation-driven heuristic for addressing order assignment and pod selection in Robotic Mobile Fulfillment Systems (RMFS), specifically focusing on a single picker station environment. Their approach begins by grouping orders into batches using a column generation framework. Furthermore, Cheng et al. [56] focus on improving order batching in Robotic Mobile Fulfillment Systems (RMFS) by introducing a deep reinforcement learning (DRL) hyper-heuristic framework. This work contributes primarily to the order-side optimization of RMFS and showcases the potential of machine learning in adaptive fulfillment operations.

Wu et al. [32] present a comprehensive five-stage mixed-integer programming model that jointly optimizes order picking and replenishment in RMFS. Their model strategically coordinates order and pod assignment, robot deployment, route planning, and replenishment operations. Additionally, related research highlights the significance of storage location assignment and SKU correlation in enhancing POA performance. Zhang et al. [21] developed a model that leverages item demand correlations to minimize travel distances, using a simplified MINLP formulation. Similarly, Shi et al. [57] address large-scale order picking optimization in RMFS with a two-stage method: the first identifies a minimal set of critical pods, and the second uses a Marginal Return-based Assignment to simultaneously assign orders and pods to stations.

In the context of online and real-time decision-making, Jiao et al. [7] proposed an online joint optimization model for Pick Order Assignment (POA) and Pick Pod Selection (PPS), incorporating alternating order-driven heuristics within a simulation framework. In contrast, Xie et al. [26] introduced a Mixed-Integer Programming (MIP) model that

assumes all orders are known in advance by aggregating those received within the past hour. Their approach combines POA and PPS decisions and further extends the model by allowing order splitting, marking the first study to explicitly address split orders in this context.

Despite notable progress in RMFS research, many existing models focus on minimizing pod-station visits or optimizing individual pod usage. However, these local optimizations do not always translate into higher system-wide throughput. In contrast, this study prioritizes throughput maximization by introducing a Future-Aware order assignment. This approach dynamically evaluates inventory conditions and anticipated bin availability across all picking stations, enabling the system to assign orders to picking stations with the highest fulfillment potential. By aligning the forward-looking order assignment with picking station demand-driven pod selection, the framework prioritizes pods that can fulfill multiple order lines in a single visit. This results in a more coordinated and adaptive RMFS, capable of maintaining high throughput, reducing order completion time, and improving pile-on performance under fluctuating demand conditions.

2.4 Pick Pod Selection

Pick Pod Selection (PPS) plays a critical role in Robotic Mobile Fulfillment Systems (RMFS), as it determines which inventory pods should be retrieved to fulfill assigned orders. The quality of pod selection directly affects picking efficiency, pod-station traffic, and robot utilization. An effective PPS strategy can minimize redundant pod movements, reduce congestion, and enhance overall system throughput.

Yang et al. [29] introduced a mixed integer linear programming model and proposes a two-stage approach to optimize order scheduling and pod sequencing within the RMFS, aiming to reduce the total number of pod visits required for order fulfillment. Wang et al. [28] addresses the combined optimization of order sequencing and rack scheduling across multiple picking stations, allowing pods to visit more than one station in succession to fulfill additional orders. The objective is to minimize the overall number of pod movements required during the fulfillment process. Consistently, Teck and Dewil [30] address pod selection as part of a broader set of RMFS decisions, including order assignment and robot dispatching. They develop integrated MIP models and incorporate interstation travel, allowing pods to move directly between pick stations without returning to storage.

Zhuang et al. [58] investigate the optimization of rack configuration and pod storage assignment within Robotic Mobile Fulfillment Systems (RMFS) to enhance picking efficiency, by reduce the rack movements. Their study focuses on long-term planning based on historical order data, where similar items are grouped on pods to reduce retrieval trips and pod assignment, where pods are selected to picking stations based on SKU turnover rate, product correlation, and purchase frequency of products.

Zou et al. [40] further investigated intelligent assignment rules to balance robot workloads and pod usage, showing that efficient pod selection contributes to decongesting pick station areas. Similarly, Krenzler et al. [46] developed deterministic selection heuristics that consider inventory availability and picker workload balancing.

Zhuang et al. [48] address the rack retrieval and repositioning problem in Robotic Mobile Fulfillment Systems (RMFS) with the objective of optimizing pod movement efficiency. Their work focus on minimizes system makes-pan by intelligently sequencing both the retrieval and repositioning of pods across multiple stations.

Recent literature also highlights the importance of connecting PPS with other operational decisions, such as replenishment planning and order SKU grouping. Wu et al. [32] illustrated how integrating PPS with replenishment processes can yield system-wide performance gains. Kim et al. [23] showed that grouping frequently co-ordered SKUs into the same or nearby pods reduces travel and enhances pile-on, using a heuristic and optimization approach to maintain item similarity under order demand.

Early Pick Pod Selection (PPS) strategies primarily focused on minimizing the number of pod-to-station visits. While effective in reducing pod traffic, this objective alone may not fully capture the operational efficiency of Robotic Mobile Fulfillment Systems (RMFS). A more critical performance metric is pile-on, defined as the average number of SKUs retrieved from a pod during a single station visit. Higher pile-on values indicate that more order lines are fulfilled per trip, contributing to reduced robot travel and improved order picking process.

However, static models that optimize pod selection based on fixed rules often struggle to adapt to the dynamic nature of Robotic Mobile Fulfillment Systems (RMFS). In practice, real-time fluctuations in inventory levels and bin availability require more flexible and responsive strategies.

To address these challenges, this study introduces a real-time, demand-aware pod selection method. The proposed approach evaluates SKU demand at each picking station and scores candidate pods based on their potential to fulfill multiple order lines across current and pre-assigned orders. By prioritizing pods with higher fulfillment potential, the strategy aims to maximize pile-on and improve system throughput, while maintaining high responsiveness and operational accuracy.

2.5 Research Gap and Contribution

Despite notable progress in operational decision-making for Robotic Mobile Fulfillment Systems (RMFS), several critical gaps remain in the existing literature, as highlighted and addressed in this study. First, many prior studies have been developed under static assumptions, such as fixed order arrivals, unchanging inventory levels, and pre-assigned pods. Second, several existing studies adopt overly simplified problem settings, such as modeling only a single picking station, assuming single-line orders, and neglecting replenishment operations. Third, prior approaches often lack mechanisms for coordinating decisions across orders, picking stations, and pod inventories, specifically in anticipation of near-future system changes, such as projected bin availability and fluctuating pod inventory levels. These assumptions fail to capture the dynamic and complex nature of e-commerce warehouse environments, which are characterized by demand variability, SKU diversity, dynamic pod movements, multiple picking stations, multi-line orders, and ongoing replenishment. Consequently, existing models often rely on static decision-making strategies that lack the responsiveness required for real-time, adaptive operations.

To address these limitations, this study proposes a Future-Aware Pick Order Assignment strategy, in which orders are dynamically assigned based on both current and anticipated bin availability across all picking stations, along with the SKU fulfillment of active pods. This approach supports many-to-many assignment scenarios, enabling multiple orders to be matched with multiple stations under forward-looking system conditions. Furthermore, a demand-driven pod selection strategy is introduced, which accounts for the total unfulfilled SKU demand from both active and pre-assigned orders at each station. Pods are then selected based on their capacity to fulfill multiple order lines simultaneously, thereby increasing the number of items picked per pod visit and enhancing overall operational efficiency.

Table 2.2 Comparison between the related literature and this study

No	Reference	Focus	Dynamic	Multiple Picking Station	Solution Method
1	Xie et al. (2021)[26]	Focus on POA and PPS by introducing split orders	✓	✓	MIP, Simulation
2	Valle and Beasley (2021)[55]	Focus on order assignment, rack assignment, and rack sequencing		✓	MIP, Heuristic
3	Zhuang et al. (2022)[31]	Order picking optimization with rack-moving, by considering rack conflicts.		✓	MIP, Heuristic
4	Wang et al. (2023) [28]	Joint optimization of order and rack sequencing in systems with multiple picking stations.		✓	Heuristic
5	Jiao et al. (2023) [7]	Alternative execution of POA and PPS through online joint optimization	✓	✓	MIP, Heuristic
6	Justkowiak and Pesch (2023) [33]	Order scheduling and rack sequencing using column generation heuristic.			Heuristic
7	Cheng et al. (2024) [56]	Order batching using DRL hyper-heuristics.	✓	✓	DRL
8	Shi et al. (2024) [57]	Fulfillment-focused simultaneous assignment for large-scale warehouse.		✓	Heuristic
9	Yang et al. (2025) [59]	Joint optimization of order scheduling and rack scheduling			MILP, Heuristic
10	Wu et al. (2025) [32]	Joint optimization of order picking and replenishment in RMFS.		✓	VNS
11	This Study (2025)	Dynamic POA and PPS using future-aware with real-time inventory.	✓	✓	Heuristic, Simulation

CHAPTER 3

METHODOLOGY

3.1 Research Framework

This study proposes a simulation-based framework designed to enhance order picking efficiency in Robotic Mobile Fulfillment Systems (RMFS). The framework focuses on two core decision processes: Pick Order Assignment (POA) and Pick Pod Selection (PPS). The proposed framework is designed to address the operational challenges posed by high order volumes and diverse SKUs. It dynamically adapts to fluctuations in inventory levels and the operational status of picking stations, ensuring that real-time decisions support high-throughput performance.

The first component, Pick Order Assignment (POA), determines the optimal assignment of unassigned orders to available picking stations. This decision considers both current and anticipated inventory conditions by evaluating the SKU fulfillment potential based on the active pods associated with each picking station. By prioritizing assignments with higher fulfillment rates, the POA mechanism aims to improve the efficiency of picking process and reduce order completion times.

The second component, Pick Pod Selection (PPS), is responsible for identifying the most suitable pod to dispatch to a picking station. Rather than selecting pods based solely on individual order requirements, the proposed PPS strategy aggregates SKU demand from all active orders at the picking station and selects the pod capable of fulfilling the highest number of required SKUs. This demand-oriented approach is designed to maximize pile-on and improve overall system efficiency.

The order assignment and pod selection modules operate within a simulation-based framework that captures real-time warehouse dynamics to inform decision-making processes. The approach facilitates intelligent coordination of resources and enables adaptive fulfillment operations under dynamic and high-density warehouse conditions.

3.1.1 Simulation and Implementation

In this phase, the proposed decision-making framework is implemented within a discrete-event simulation environment. The simulation setup models the complete RMFS operation, including order generation, SKU-to-pod assignment, pod dispatching, robot movements, and picking operations.

The simulation utilizes tools including NetLogo, Python, and Excel to replicate warehouse dynamics under a range of operational conditions. Both the proposed framework and a baseline approach representing conventional RMFS logic are encoded for comparative analysis.

3.1.2 Performance Evaluation and Analysis

The phase to evaluate the effectiveness of the proposed framework relative to the baseline strategy using key performance indicators such as throughput, order completion time, and pile-on rate. Statistical analysis is conducted to evaluate the significance of the results and to ensure that observed differences are meaningful and consistent. In addition, a sensitivity analysis is performed to test the model's robustness under varying operational parameters, including number of robots, the number of pods, and the number of picking stations. These findings provide insights into the trade-offs and scalability of the proposed methods.

3.2 RMFS Operational Flow

In a Robotic Mobile Fulfillment System (RMFS), the operational flow involves a continuous cycle of interactions among orders, pod inventories, autonomous mobile robots (AMRs), storage areas, and picking stations. The system dynamically coordinates these elements to sustain high throughput while managing real-time constraints such as bin capacity, pod inventory, and robot availability. This study models the entire flow in a simulation environment to evaluate decision-making strategies for Pick Order Assignment (POA) and Pick Pod Selection (PPS).

The process begins with the arrival of orders, each specifying a set of SKU requirements. The system simultaneously evaluates the current availability of empty bins at each picking station and monitoring the near-future bin availability.

When bin availability is detected, the system dynamically evaluates all unassigned orders by calculating their fulfillment rate, defined as the proportion of required Stock Keeping Units (SKUs) that can be immediately fulfilled by pods currently active at the picking station. Active pods comprise those already positioned at the picking station, those in queue, and those currently in transit. This logic allows for partial fulfillment, meaning orders can be assigned even if not all SKUs are covered immediately, provided they are matched to the picking station with the highest fulfillment rate. When a bin is about to become available at another picking station, and that picking station offers higher fulfillment potential, the order is pre-assigned to that picking station. This forward-looking assignment leverages projected bin availability to support a globally optimized order flow.

Following order assignment, the system proceeds with Pick Pod Selection (PPS). Candidate pods are filtered based on the station's SKU demand profile, requiring at least one matching SKU, are then evaluated using a heuristic that combines the SKU match rate, which indicates how well the pod satisfies the station's remaining demand with inventory quantity. Pods that can fulfill multiple order lines simultaneously are prioritized to maximize the pile-on rate and improving throughput efficiency. The selection process accounts for inventory distribution, queue capacity, and station constraints to ensure balanced pod utilization. Once a pod is selected, a corresponding task is generated for an AMR to retrieve the pod from storage and deliver it to the designated picking station.

When an AMR is assigned to retrieve a pod, its travel path is determined by considering its current location, the distance to the pod, and potential congestion within the warehouse grid. A modified Dijkstra path planner is used to calculate the most efficient route while avoiding conflicts with other robots. Robot retrieves the selected pod from storage and transports it to the designated picking station for item retrieval.

After the picking operation is completed, each pod must return either to storage or to a replenishment station before being reassigned. A pod is directed to replenishment if the inventory level of any contained SKU falls below the predefined replenishment threshold.

The system applies a standard nearest-location return rule, directing each robot to return the pod to the closest available storage slot dynamically selected according to real-time slot availability. Once a return is completed, the available robot receives the next

dispatch task, ensuring that pods are continuously brought to the appropriate picking stations in response to station requirements

Routing behavior differs depending on pod carriage status. When loaded with a pod, the robot must follow aisle-only routing due to physical clearance limits and safety protocols. Once the pod is returned, the robot gains under-rack traversal capability, enabling shorter, more direct paths to its next assignment. Furthermore, unladen robots are capable of operating at higher speeds, as they are not restricted by the weight and stability requirements associated with pod transport. By returning pods to the nearest available storage location, the system enhances overall routing flexibility and increases operational responsiveness, enabling robots to reach subsequent tasks more quickly and efficiently. Table 3.1 summarizes key differences in routing constraints during the pod return and pickup phases.

Table 3.1 Comparison of robot movement phases: pod return vs. pod pickup

Scope	Pod Return	Pod Pick up
Process	Returns the pod back to storage	Picks up the next assigned po
Initial Location	Picking station or Replenishment	Empty Location
Destination	Nearest empty storage location	Location of the selected pod
Carry Pod	Yes	No
Routing	Aisle-only routing	Aisle and Under-rack routing

3.3 Proposed Decision Framework for the Picking Process

This study proposes a simulation-based decision framework that models two key operational processes within a dynamic Robotic Mobile Fulfillment System (RMFS): Pick Order Assignment (POA) and Pick Pod Selection (PPS). The framework is designed to operate under continuously changing system states, including real-time fluctuations in inventory levels, dynamically arriving orders, and bin availability at picking station. The primary objective is to enhance picking efficiency. To achieve this, the Pick Order Assignment (POA) model is designed to maximize the number of order SKUs that can be fulfilled using the available pod inventory at each picking station. In parallel, the Pick Pod Selection (PPS) model aims to maximize the pile-on rate by selecting pods that best match the aggregated SKU demand across active orders at a picking station.

The framework continuously monitors bin status to identify availability, order status to assess completion progress, and the inventory of pods, including those that are queued, in transit, and assigned, to match orders and pods according to picking station SKU demand. This dynamic decision process allows the RMFS to adapt to variable order arrival patterns and inventory constraints. Figure 3.1 illustrates the system flowchart for the model, while Table 3.2 Notation and symbol definitions for RMFS decision modules summarizes the symbols and notations used throughout this section for clarity and consistency.

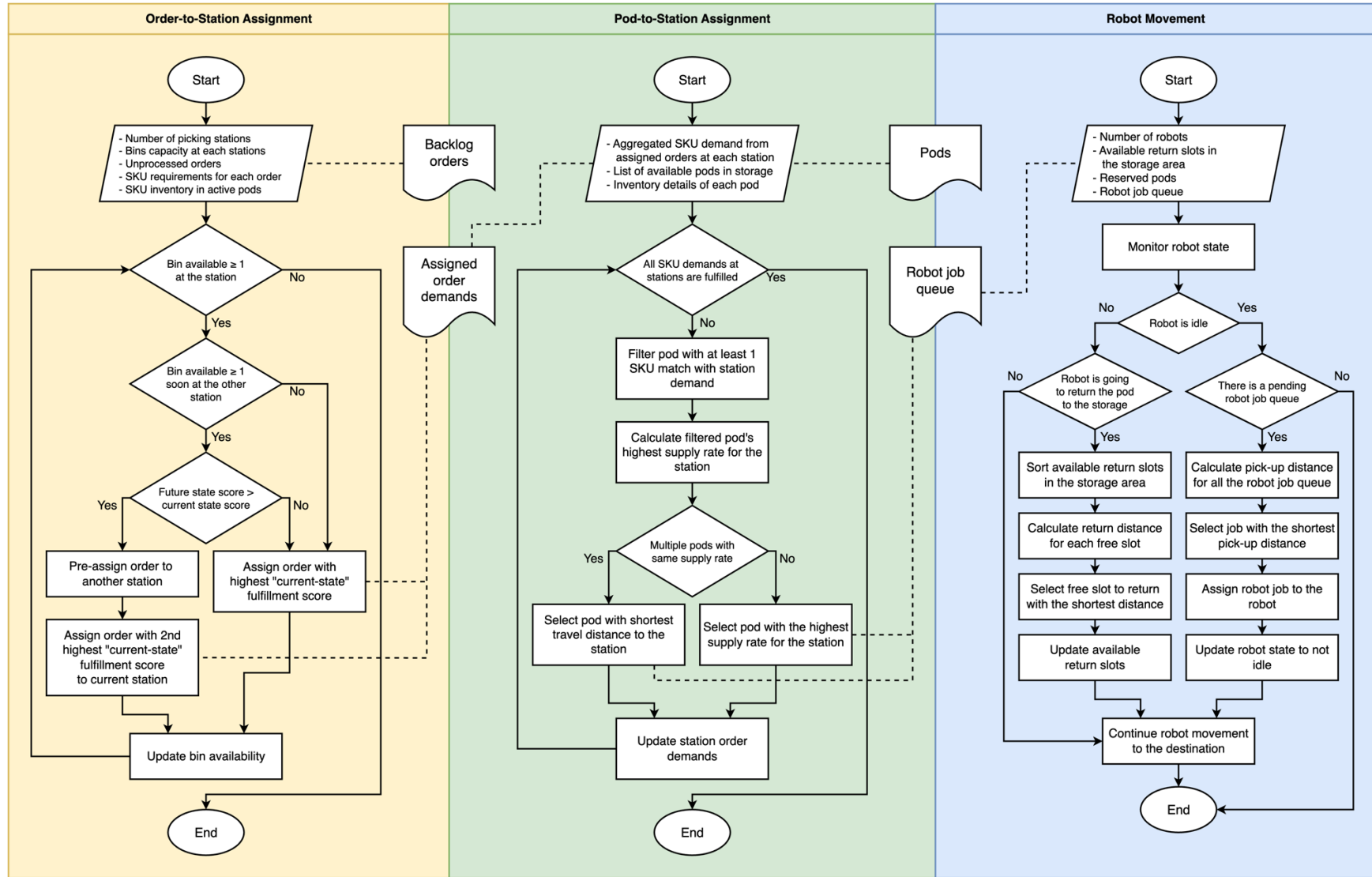


Figure 3.1 System flowchart for proposed model

Table 3.2 Notation and symbol definitions for RMFS decision modules

Notation		Description
Indices	i	Index of order
	j	Index of SKU
	p	Index of pod
	s	Index of the picking station
Parameters	C	: Capacity of each picking station
	J	: Set of all SKU
	M	: Total number of picking stations
	P	: Total number of pods
	D_{js}	: Total demand of SKU j across all orders in O_s
	SKU_i	: Set of SKUs required by order i
	P_s^{cand}	: Candidate pods considered for assignment to picking station s
	P_s	: Set of active pods currently serving picking station s
	Inv_{jp}	: Inventory level of SKU j in pod p
	O_{ij}	: Required quantity of SKU j in order i
	O_s	: Set of orders assigned to picking station s
	OB	: Set of unprocessed orders
	SKU_{ps}	: Set of SKUs needed at picking station s that are also in pod p
	$Q_{j,o,p,s}$	: Quantity of SKU j from pod p for order o at picking station s
Decision	W_{ps}	: {1, if pod is assigned to picking station s ; 0 otherwise}
Variable	X_{is}	: {1, if order o is assigned to picking station s ; 0 otherwise}

3.3.1 Pick Order Assignment

Orders are dynamically assigned to picking stations based on SKU fulfillment rates calculated using real-time inventory information from the active pod set, which is defined as the collection of pods that have been assigned, are currently queued, or are in transit to a picking station. This study proposes a Future-Aware order assignment strategy. This strategy actively monitors both current and near-future bin availability across all picking stations and dynamically matches each order to the picking station that offer the highest fulfillment potential.

In this process, the system continuously tracks bin occupancy at each picking station. The assignment module evaluates all unassigned orders by calculating the current fulfillment rate, defined as the number of each order's required SKUs that can be immediately satisfied by active pods set, which include pods already queued for processing, those in transit, and assigned. By evaluating how the required quantities of each order's SKUs can be satisfied by the real-time inventory available in the active pod set, the system prioritizes orders that achieve a higher proportion of unit-level fulfillment.

Beyond the current system state, the strategy includes a Future-Aware mechanism that estimates the near-future fulfillment rate for each station by projecting which occupied bins are going to become available soon. A bin is considered available under this mechanism if a pod is currently in the picking stage and is in the process of fulfilling the remaining SKUs required to complete the ongoing order. Such bins are considered to be available soon, enabling the anticipatory assignment of orders. As illustrated in Figure 3.2, the system continuously monitors all picking stations to detect current available bins and those anticipated to become available in the near future.

The model evaluates all candidate bins to estimate the fulfillment potential for each order. At the projected future time when a bin becomes available, the system evaluates the fulfillment potential of each order based on the inventory that will be accessible at that moment, as illustrated in Figure 3.3. This evaluation is based on the projected inventory of the next active pod set, ensuring that the estimated fulfillment rate accurately reflects future inventory conditions. If an alternative station is projected to offer a higher fulfillment rate in this near-future state, the order is pre-assigned to that station. In parallel, a different order is selected and assigned to the currently available bin based on the current fulfillment rate of that station's inventory, as illustrated in Figure 3.4. The forward-looking logic selects order-to-station assignments that maximize the number of SKUs fulfilled by the active pod inventory. The objective of the Pick Order Assignment (POA) model is to maximize the number of order SKUs that can be fulfilled using this active inventory.

$$FR_{i,s} = \frac{\sum_{j \in SKU_i} \min(\sum_{p \in P_s} Inv_{jp} > 0)}{|SKU_i|}, \forall_i \in OB \quad (3.1)$$

$$\max \sum_{i \in OB} \sum_{s \in S} FR_{i,s} \cdot x_{i,s} \quad (3.2)$$

$$\sum_{s \in S} x_{i,s} \leq 1, \quad \forall_i \in OB \quad (3.3)$$

$$\sum_{i \in O} x_{i,s} \leq C, \quad \forall_s \in S \quad (3.4)$$

$$x_{i,s} \in \{0,1\}, \quad \forall_i \in O, s \in S \quad (3.5)$$

$FR_{i,s}$ is the fulfillment rate is evaluated for each unassigned order $i \in OB$ based on the quantity of SKU inventory levels of pods assigned, queued, and in transit to the picking station, as shown in (3.1). In this framework, the fulfillment rate $FR_{i,s}$ uses two related pod sets depending on the assignment scenario. For real-time assignment, the active pod set $P_s^{current}$ includes pods that have been assigned, are queued, and are in transit to station s . For pre-assignment, the projected pod set P_s^{future} represents the inventory that will be available at the station once the bin becomes accessible, reflecting any changes in pod presence due to ongoing picking operations. This distinction ensures that the calculated fulfillment rate reflects the available quantities of required SKUs at the time of order assigned.

The objective function (3.2) maximizes the sum of fulfillment scores across all combinations of unassigned orders and available stations. The model prioritizes assignments that offer the highest SKU coverage with existing pod resources. To ensure logical feasibility, Constraint (3.3) restricts each unassigned order to be assigned to at most one picking station. Constraint (3.4) limits the number of assigned orders at each station to not exceed its processing capacity C , which may correspond to the number of available bins. Constraint (3.5) declares the binary nature of the decision variable $x_{i,s}$, which takes a value of 1 if order i is assigned to station s , and 0 otherwise. When multiple orders yield identical fulfillment scores, the system applies an arrival-time priority rule, ensuring that earlier orders are assigned first.

To illustrate the calculation of the unit-level fulfillment rate defined in (3.1), consider an unassigned order i that requires three SKUs: 2 units of SKU A, 1 unit of SKU B, and 4 units of SKU C. The total required quantity for the order is 7 units. Suppose the active pod set P_s at picking station s includes two pods with the following inventory levels: Pod 1 contains 1 unit of SKU A and 2 units of SKU C, while Pod 2 contains 1 unit of SKU A, 1 unit of SKU B, and 1 unit of SKU C. The aggregated inventory across the active pods is 2 units of SKU A, 1 unit of SKU B, and 3 units of SKU C. The contribution of each SKU is limited to its required quantity using the minimum function. In this example, the fulfilled quantity is calculated as the minimum between the total available inventory from the active pod set and the required quantity for each SKU. The minimum value for SKU A is 2, for SKU B is 1, and for SKU C is 3, resulting in a total fulfilled quantity of 6 units. The total required quantity for the order is 7 units. Therefore, the resulting fulfillment rate is approximately 0.857.

This Future-Aware Order Assignment strategy, driven by SKU fulfillment rates, enables adaptive load balancing across picking stations and continuously aligns the order flow with dynamic pod movements. By jointly considering current and anticipated system states, this approach maintains high picking efficiency, reduce order completion time, and increases pile-on opportunities, and improves overall RMFS performance in dynamic, high-SKU environments.

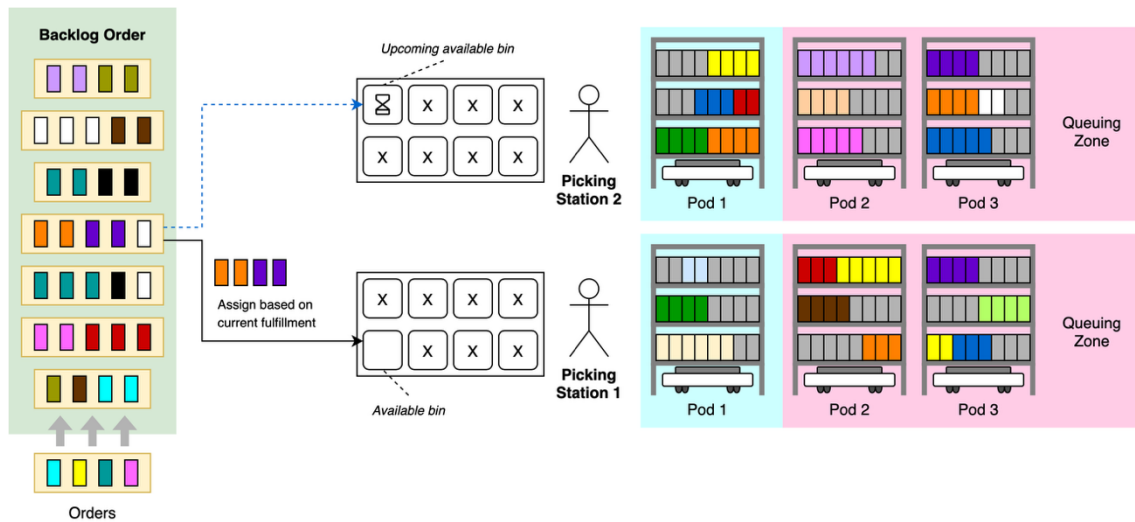


Figure 3.2 Detection of bin availability

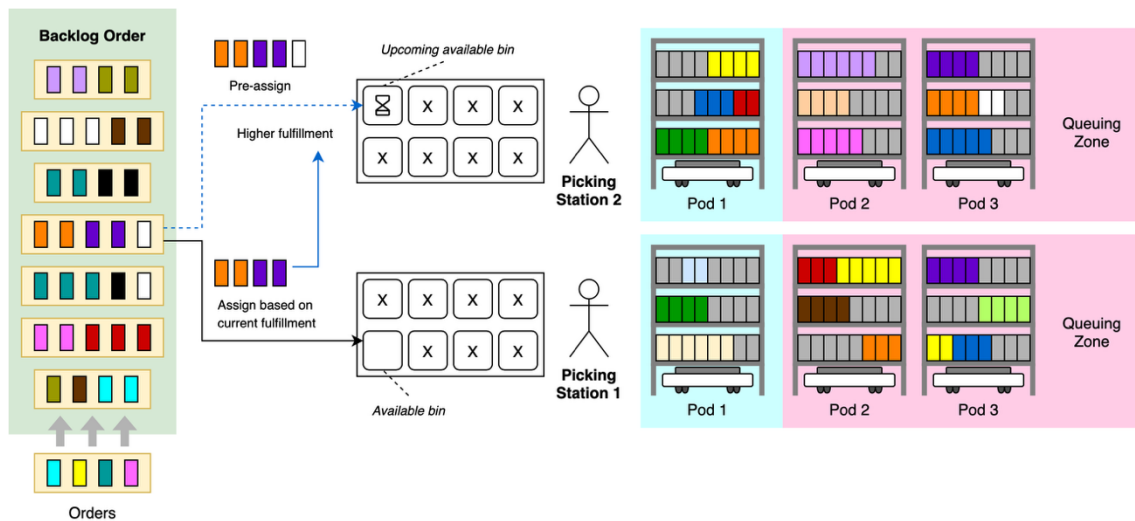


Figure 3.3 Cross-station fulfillment evaluation

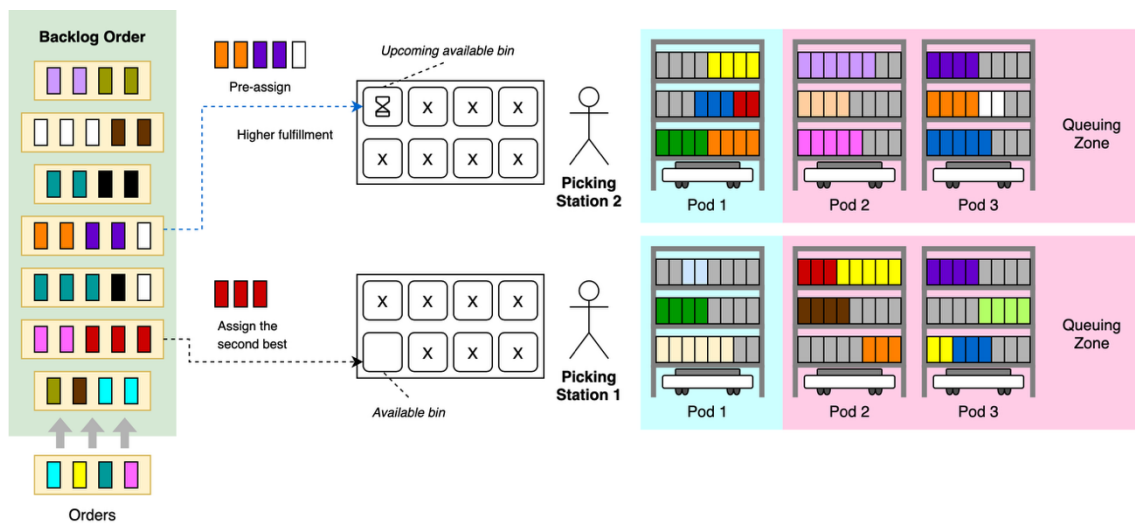


Figure 3.4 Future-aware order assignment

3.3.2 Pick Pod Selection

The Pick Pod Selection (PPS) module aims to increase picking process by selecting inventory pods that maximize SKU supply alignment with the requirements of each picking station. The proposed approach adopts a Station Match Strategy, which prioritizes pods based on their inventory level to supply the unfulfilled SKU quantities of orders currently assigned to a given picking station, including both assigned and pre-assigned orders determined by the Pick Order Assignment module.

The Pick Pod Selection (PPS) module operates in two primary stages: candidate filtering and pod evaluation. This process begins with the system collecting the unfulfilled SKU requirement for each picking station s . A pod is included in the candidate set P_s^{cand} if it is available in the storage area, meaning it is not currently queued, in transit, or already assigned to any picking station, and contains at least one SKU that matches the unfulfilled SKU demand from all pre-assigned orders at picking station s . Each candidate pod is evaluated based on its contribution to supply the picking station's demand. The system calculates a supply score for each pod by summing the quantity of SKUs that align with the picking station's current demand, as defined in Equation (3.6) and illustrated in Figure 3.5. Pods are then ranked according to this score, reflecting how well their inventory aligns with demand of active orders at the picking station. The system selects the pod with the highest score. If multiple pods yield the same fulfillment score, the system applies a tie-breaking rule by selecting the pod that is closer to the picking station. The proximity is measured using the Manhattan distance, calculated the differences in the horizontal and vertical coordinates between the pod's current location and the target picking station.

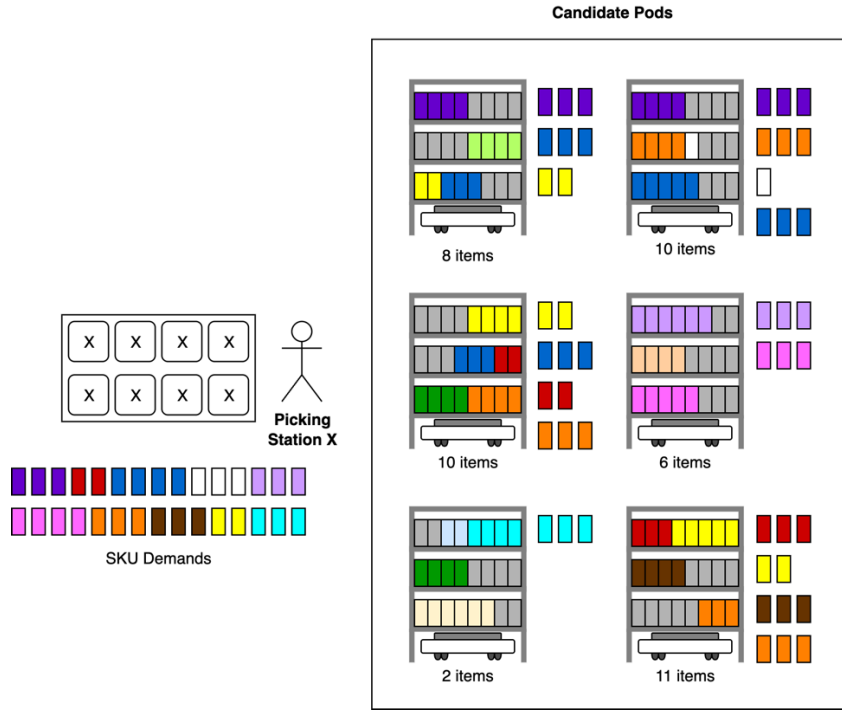


Figure 3.5 Station demand driven pod selection

The primary objective of the Pick Pod Selection (PPS) module is to maximize the number of item quantities that can be picked from a selected pod, based on the current demand at each picking station.

$$\max \sum_{p \in P_s^{cand}} W_{p,s} \left(\sum_{j \in SKU_{ps}} \min(Inv_{pj}, D_{js}) \right) \quad (3.6)$$

$$\sum_{p \in P} W_{p,s} \leq 1, \quad \forall_s \in S \quad (3.7)$$

$$\sum_{i \in O_s} Q_{j,i,p,s} \leq Inv_{pj} \cdot W_{p,s}, \quad \forall_j \in J, p \in P, s \in S \quad (3.8)$$

$$W_{p,s} \in \{0,1\}, \quad \forall_p \in P, s \in S \quad (3.9)$$

To guide pod selection in fulfillment operations, the objective function in (3.6) aims to select a pod that maximizes the number of SKU units it can contribute to meeting the total demand of a picking station. For each candidate pod, the system evaluates its potential by summing the minimum values between the pod's inventory for each SKU j and the total demand D_{js} at picking station s . Constraint (3.7) ensure that each pod can

be assigned to at most one picking station. Constraint (3.8) enforces inventory validity by ensuring that the total quantity of SKU j supplied by pod p across all relevant orders does not exceed the pod's stored inventory level for that SKU. Finally, Constraint (3.9) ensures that the decision variable $W_{p,s}$ is binary, where $W_{p,s} = 1$ indicates that pod p is assigned to picking station s , and $W_{p,s} = 0$ otherwise.

The following example outlines how the system evaluates and selects a pod. Suppose Picking Station s has total unfulfilled SKU for three SKUs: SKU A requires 4 units, SKU B requires 2 units, and SKU C requires 3 units. After filtering out pods, two candidate pods remain: Pod P1 and Pod P2. Pod P1 contains 3 units of SKU A, 2 units of SKU B, and 1 unit of SKU C, while Pod P2 contains 4 units of SKU A, 1 unit of SKU B, and 2 units of SKU C. The system applies the minimum function to each SKU by comparing the available inventory in the pod with the total demand at the picking station. For Pod P1, the fulfilled quantity is computed as 3 units for SKU A, 2 units for SKU B, and 1 unit for SKU C, resulting in a total of 6 fulfilled units. In comparison, Pod P2 fulfills 4 units for SKU A, 1 unit for SKU B, and 2 units for SKU C, yielding a total of 7 fulfilled units. As Pod P2 contributes more effectively to meeting the total SKU demand at the picking station, it is selected for dispatch.

The Pick Pod Selection (PPS) module plays a critical role in matching station demand with available pod inventory. Candidate pods are filtered and ranked to ensure that each dispatched pod contributes meaningfully to active picking tasks. This demand-driven approach supports throughput performance and enhances pile-on efficiency in dynamic warehouse environments.

3.3.3 Baseline Strategy and Decision Rule

To evaluate the effectiveness of the proposed decision framework, a comparative analysis is conducted against a baseline strategy that is widely referenced in existing RMFS literature. Following the approach described by Merschformann et al. [12], the baseline configuration follows two heuristic rules known to perform well under operating conditions. These rules represent a combination of order assignment (POA) and pod selection (PPS) strategies that have achieved best throughput score. Table 3.3 summarizes the decision rules applied in the simulation experiments for both the baseline and the proposed models.

For Pick Order Assignment (POA), the Pod-Match rule is applied. Under this rule, orders are selected from the backlog based on their alignment with the pods already heading to the picking station. For Pick Pod Selection (PPS), the Demand rule is selecting the pod whose contents are most demanded based on the current order backlog situation.

In addition to Pick Order Assignment (POA) and Pick Pod Selection (PPS). The simulation framework includes several other operational-level decisions that are handled using standardized rules. These include Replenishment Order Assignment (ROA), Replenishment Pod Selection (RPS), Task Allocation (TA), Path Planning (PP), and Pod Relocation (PR). Path planning relies on Dijkstra's algorithm with Manhattan distance-based cost calculations; this method supports efficient and grid-compliant navigation in RMFS layouts.

Table 3.3 Decisions rules in the simulation experiment performance analysis

Decisions Problem	Baseline	Proposed
POA	Pod-Match	This research
PPS	Demand	This research
ROA	Random	Random
RPS	Emptiest	Emptiest
TA	Nearest	Nearest
PR	Nearest	Nearest

3.4 System Configuration

The Robotic Mobile Fulfillment System (RMFS) employed in this study is designed to emulate a realistic warehouse environment, integrating simulation, data processing, and optimization tools. This section presents the technical architecture of the system and the physical layout configuration used to implement and evaluate the proposed decision-making framework.

3.4.1 System Architecture

A comprehensive and integrated simulation environment was developed using NetLogo, Python, and Microsoft Excel to support the modeling, optimization, and analysis of RMFS operations. The system architecture, as illustrated in Figure 3.6, facilitates dynamic communication across these platforms to simulate warehouse workflows and implement optimization strategies.

NetLogo serves as the primary simulation engine, providing a visual and agent-based environment for modeling the warehouse. In this platform, agents represent key physical and operational components of the RMFS, including storage pods, Automated Mobile Robots (AMRs), picking stations, and replenishment zones. NetLogo's interactive simulation capabilities allow for real-time visualization and behavioral tracking of warehouse entities, enabling effective observation of dynamic system performance.

Python is used as the core programming tool for executing complex decision logic and real-time control of the simulation. It handles order assignments, pod selection, robot task coordination, and fulfillment calculations. Through direct communication with NetLogo, Python processes optimization routines and updates the simulation state based on changing system conditions, ensuring dynamic and adaptive decision-making.

Microsoft Excel functions as the data management interface within the system. It stores essential inputs such as layout parameters, inventory levels, order lists, and SKU distributions, and receives output data for performance evaluation. Excel enables flexible data manipulation by importing and exporting CSV files from NetLogo and Python, facilitating rapid adjustments to simulation parameters and supporting scenario-based analysis.

The typical workflow begins with setting operational parameters and warehouse layouts in Excel. These configurations are imported into NetLogo, where the simulation is executed. Throughout the simulation, Python scripts dynamically manage task allocation and optimization logic. Upon completion, performance data is exported to Excel for post-simulation reporting and analysis.

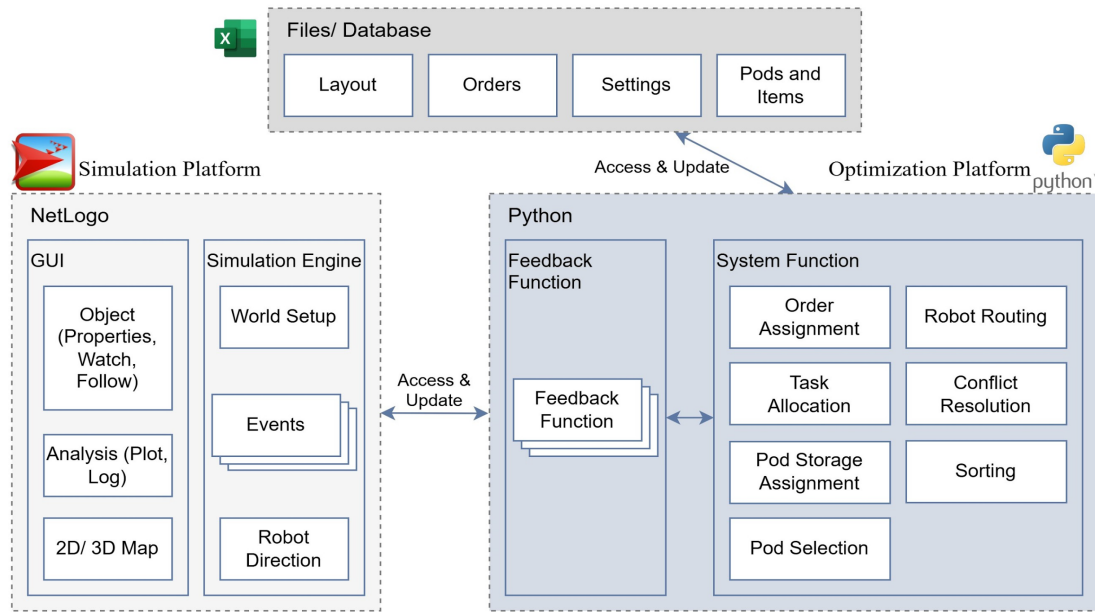


Figure 3.6 System architecture

3.4.2 Simulation Layout

The warehouse layout implemented in NetLogo is shown in Figure 3.7. It comprises three functional zones: picking stations, storage areas, and replenishment stations.

- Picking stations, located on the left side of the layout, are staffed with human pickers and supported by queue lanes for AMRs. Each picker transfers SKUs from pods to dedicated bins to fulfill orders. Each bin serves one order at a time, reflecting the picking station's limited processing capacity.
- The central storage area houses the pods containing SKUs. Unoccupied grid cells represent vacant storage positions, which serve as potential pod return locations. AMRs may return pods to these locations once a picking or replenishment task is complete.

- Replenishment stations are positioned on the right side of the layout. When inventory levels for a specific SKU fall below a predefined threshold, affected pods are routed here. AMRs facilitate this process by transporting pods to the replenishment stations, where stock is replenished before returning to storage.

AMRs operate within one-way aisles and can move bidirectionally along straight paths or turn at junctions to navigate the warehouse. Dual-lane corridors adjacent to the picking and replenishment zones provide multiple access routes, supporting efficient task flow and minimizing congestion. In addition, charging is necessary when robots run low on batteries. However, since this study mainly focuses on POA and PPS at the operational level, the battery power of all robots in Netlogo is infinite, i.e. the robot in Netlogo does not need to charge.

To support real-time monitoring and debugging during simulation, AMRs are color-coded to indicate their current status:

- Black: Idle, awaiting task assignment
- Green: In transit to retrieve an assigned pod
- Red: Transporting a pod to a picking station
- Blue: At a picking station, either undergoing or awaiting item picking
- Yellow: Transporting the pod back to storage after picking
- Pink: Assigned to a replenishment task due to low inventory

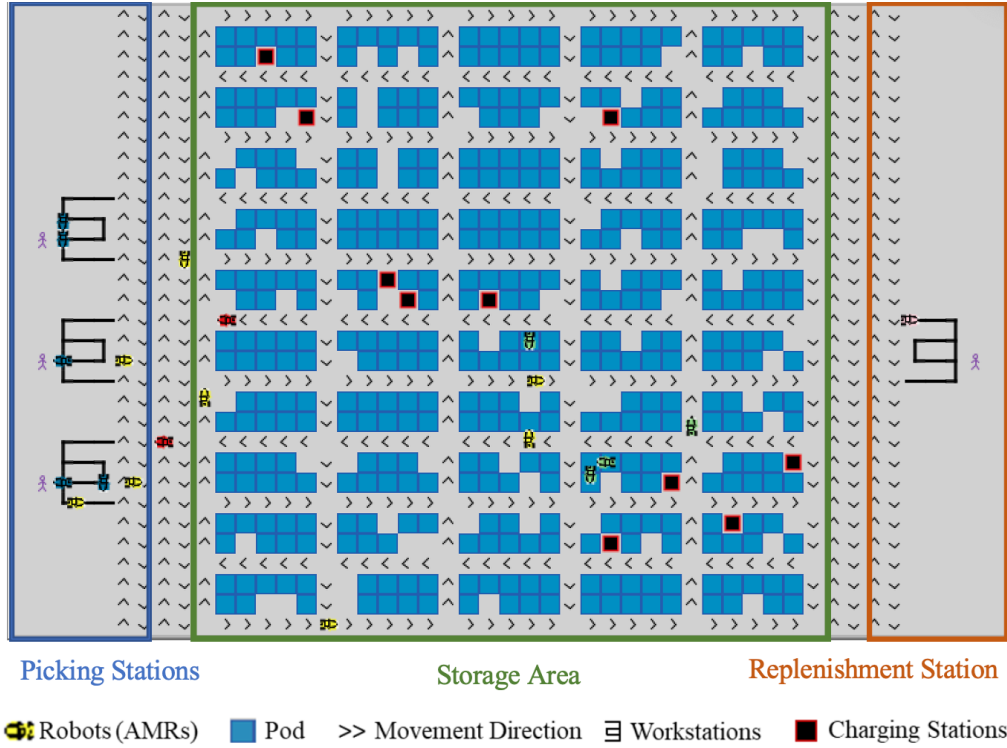


Figure 3.7 Simulation layout

3.4.3 Simulation Parameter

This study defines a set of simulation parameters to ensure that the proposed multi-stage decision framework is tested under realistic warehouse operating conditions. The simulation model replicates key aspects of a Robotic Mobile Fulfillment System (RMFS), including order arrivals, SKU distribution, pod configurations, picking station capacities, and robot fleet size.

To maintain consistency and focus on order picking efficiency, it is assumed that all orders are fulfilled within a single operational day. Accordingly, the model does not incorporate lateness penalties or priority-based scheduling. Furthermore, given pod capacity and dimensional constraints, a single pod may be insufficient to fulfill an entire order, necessitating the use of multiple pods. These considerations reinforce the importance of coordinated and adaptive assignment strategies.

The main simulation parameters are summarized in Table 3.4, and their values are based on commonly referenced RMFS configurations in literature and practical industry benchmarks.

Table 3.4 Simulation parameters

Parameters		Value
Simulation	Run Length	8 hours
	Replication	20 for each scenario
Layout	Inventory Area	500 locations
	Inventory Capacity	420 pods
	Aisles	11 horizontal aisles
		6 vertical aisles
	Stations	3 picking stations
		1 replenishment station
		10 robot charging stations
Inventory	Total number of SKUs	500 units
	SKU Distribution	Class A: 10% Class B: 30% Class C: 60%
	Number of SKUs per Pod	10 SKUs (Mixed Class of SKU)
	Replenishment Inventory Level	Class A = 40% Class B = 50% Class C = 60%
Orders	Order line	Multi-line orders
	Initial orders	100 orders
	Order arrival rate	300 orders/hour
	Quantity of items in the order line	Truncated normal distribution
	Quantity of SKU types in the order	Geometric distribution
Stations	Picking Station Capacity	8 bins for each picking station
	Process Picking Time	15 s
	Replenishment Time	20 s
Robots	Number of robots	20 AMRs
	Robot acceleration/deceleration	1 m/s
	Robot maximize velocity	1.5 m/s
	The time needed for the full turn	2.5 s
	The time needed to lift & store the pods	2.5 s

3.5 Performance Evaluation

The performance evaluation is conducted to assess the effectiveness of the proposed decision framework in improving the operational efficiency of the Robotic Mobile Fulfillment System (RMFS). The evaluation focuses on quantifiable metrics that reflect system productivity and resource utilization. This study employs 3 key performance metrics to evaluate and compare the proposed framework and the baseline strategy. Each metric is defined below, along with its calculation method and practical interpretation.

1. Throughput (TP):

Throughput is a key performance indicator used to evaluate the operational efficiency of a Robotic Mobile Fulfillment System (RMFS). It measures the number of orders fulfilled per unit of time and is defined as:

$$TP = \frac{N}{T} \quad (3.10)$$

where N is the total number of orders fulfilled, and T is the total simulation time in hours. A higher throughput value indicates greater system productivity and responsiveness, thereby contributing to the overall operational performance of the RMFS.

2. Order Completion Time (OCT):

The Order Completion Time represents the average time required to complete an order, starting from its assigned to the picking station to the point of full completion. It is calculated as:

$$OCT = \frac{\sum_{i=1}^N T_i}{N} \quad (3.11)$$

where T_i is the completion time for order i in second, and N is the total number of fulfilled orders. A lower OCT value reflects enhanced order processing efficiency and improved service speed.

3. Pile-on (PO):

Pile-on measures the average number of items picked from a pod each time it visits a picking station. It is defined as:

$$PO = \frac{\text{Total Number of Items Picked}}{\text{Total Number of Pod – Station Visit}(PSV)} \quad (3.12)$$

A higher pile-on value signifies more efficient use of pod resources, fewer pod-station visits (PSVs), and reduced robot workload.

3.6 Statistical Analysis

To ensure statistical rigor in evaluating the effectiveness of the proposed framework, Analysis of Variance (ANOVA) was performed. ANOVA is used to determine whether there are any statistically significant differences in the mean performance metrics between the proposed and baseline strategies. The null hypothesis (H_0) assumes that there is no significant difference in the mean values of the performance metrics, while the alternative hypothesis (H_1) suggests that at least one performance metric differs significantly between the two methods. Additionally, for a non-parametric statistical test used in this study is the Kruskal–Wallis test, which evaluates whether there are statistically significant differences between the medians of two or more independent groups.

3.7 Sensitivity Analysis

In addition to the baseline performance evaluation, a sensitivity analysis is conducted to evaluate the robustness of the proposed decision-making method and to determine how changes in key input parameters affect system performance. The purpose of the sensitivity analysis is identifying the parameters that most significantly affect performance, and evaluating the model's stability when subjected to changes in system configuration.

Three operational factors are selected for sensitivity testing based on their practical significance in real-world warehouse management:

1. Number of robots varied from 10 to 40
2. Number of pods varied from 380 to 460
3. Number of picking stations varied from 2 to 5

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results of our simulation model, focusing on the performance of both the baseline and the proposed decision-making strategy encompassing Pick Order Assignment (POA) and Pick Pod Selection (PPS). The chapter concludes with a comparative discussion between the two approaches, highlighting the operational improvements and trade-offs achieved through our integrated strategy.

4.1 Experimental Result

The simulation process in this research was conducted using the NetLogo simulation platform, adhering to the specific parameter settings outlined in Section 3.3.

To comprehensively assess the impact of each decision rule, this study adopts combination design that tests all possible pairings of Pick Order Assignment (POA) and Pick Pod Selection (PPS) strategies. This factorial approach isolates the individual effects of each strategy as well as their combined impact on overall system performance. The following four scenarios were implemented:

1. *PM_DM*: Pod-Match POA with Demand PPS, serving as the baseline.
2. *PM_SM*: Pod-Match POA with Station Match PPS.
3. *FA_DM*: Future-Aware POA with Demand PPS.
4. *FA_SM*: Future-Aware POA with Station Match PPS, serving as the proposed.

The simulation results evaluate the effectiveness of the proposed decision-making framework for the Robotic Mobile Fulfillment System (RMFS). The system's performance is assessed using three key operational metrics: Throughput (TP), Order Completion Time (OCT), and Pile-on (PO).

Table 4.1 shows the throughput (TP) achieved across all scenarios POA–PPS scenarios, while Table 4.2 presents the corresponding order completion time (OCT) results. The pile-on (PO) results, which represent pod utilization efficiency, are presented in Table 4.3 for each strategy combination. Finally, Table 4.4 provides a comparative summary of the baseline configuration and the proposed configuration across all three key performance indicators: throughput, order completion time, and pile-on.

Table 4.1 Throughput performance across scenario of POA and PPS

Replication	Throughput			
	PM_DM	PM_SM	FA_DM	FA_SM
1	245	260	263	292
2	245	261	257	290
3	243	258	255	288
4	246	264	261	293
5	241	259	254	289
6	247	265	262	295
7	242	262	260	291
8	243	258	253	288
9	240	256	251	282
10	244	260	258	290
11	241	256	253	286
12	243	264	259	292
13	243	259	255	287
14	240	256	253	284
15	244	260	257	291
16	242	259	254	288
17	239	255	252	284
18	247	262	259	291
19	238	254	250	281
20	240	257	253	286
Average	243	259	256	288
Improvement		6.84%	5.48%	18.80%

Table 4.2 Order completion time across scenario of POA and PPS

Replication	Order Completion Time (Second)			
	PM_DM	PM_SM	FA_DM	FA_SM
1	328.8	322.8	314.4	304.8
2	330.6	324.6	322.8	296.7
3	329.4	324.2	310.2	297.1
4	331.2	325.2	306.7	292.8
5	335.2	328.7	318.1	290.9
6	332.1	330.8	320.5	297.5
7	334.9	327.3	313.9	294.9
8	332.5	322.5	318.6	298.8
9	340.3	336.9	319.4	297.9
10	334.1	329.1	311.9	297.4
11	341.4	328.3	307.3	299.9
12	330.4	333.6	322.2	292.9
13	322.5	323.4	316.4	288.3
14	340.6	312.5	314.8	302.2
15	340.1	330.2	321.8	294.4
16	337.8	321.8	309.3	293.7
17	334.8	325.6	315.1	301.8
18	328.8	323.4	317.4	294.9
19	340.2	324.6	309.4	267.5
20	327.3	325.8	321.6	301.6
Average	333.7	326.1	315.6	295.3
Improvement		-2.28%	-5.41%	-11.50%

Table 4.3 Pile-on rate performance across scenario of POA and PPS

Replication	Pile-on			
	PM_DM	PM_SM	FA_DM	FA_SM
1	6.1	8.1	6.6	8.5
2	6.4	8.2	7.0	8.4
3	6.3	8.3	6.9	8.6
4	6.2	8.1	6.3	8.7
5	6.3	8.5	6.9	8.3
6	6.1	8.1	6.7	8.5
7	6.4	8.3	6.8	8.4
8	6.2	8.5	6.9	8.3
9	6.3	8.4	6.6	8.5
10	6.1	8.3	6.7	8.4
11	6.2	8.1	6.8	8.6
12	6.3	8.4	6.9	8.5
13	6.4	8.0	6.6	8.2
14	6.3	8.5	6.7	8.3
15	6.3	8.0	7.1	8.4
16	6.1	8.2	6.7	8.3
17	6.2	8.3	6.5	8.2
18	6.4	8.2	6.8	8.4
19	6.1	8.4	6.8	8.3
20	6.2	8.2	6.2	8.6
Average	6.2	8.3	6.7	8.4
Improvement		32.22%	7.77%	34.86%

Table 4.4 Comparative evaluation of baseline and proposed strategies

Replication	Performance					
	Throughput		Order Completion Time (Second)		Pile-on	
	Baseline	Propose	Baseline	Propose	Baseline	Propose
1	245	292	328.8	304.8	6.1	8.5
2	245	290	330.6	296.7	6.4	8.4
3	243	288	329.4	297.1	6.3	8.6
4	246	293	331.2	292.8	6.2	8.7
5	241	289	335.2	290.9	6.3	8.3
6	247	295	332.1	297.5	6.1	8.5
7	242	291	334.9	294.9	6.4	8.4
8	243	288	332.5	298.8	6.2	8.3
9	240	282	340.3	297.9	6.3	8.5
10	244	290	334.1	297.4	6.1	8.4
11	241	286	341.4	299.9	6.2	8.6
12	243	292	330.4	292.9	6.3	8.5
13	243	287	322.5	288.3	6.4	8.2
14	240	284	340.6	302.2	6.3	8.3
15	244	291	340.1	294.4	6.3	8.4
16	242	288	337.8	293.7	6.1	8.3
17	239	284	334.8	301.8	6.2	8.2
18	247	291	328.8	294.9	6.4	8.4
19	238	281	340.2	267.5	6.1	8.3
20	240	286	327.3	301.6	6.2	8.6
Average	243	288	333.7	295.3	6.2	8.4
Improvement	18.80%		-11.50%		34.86%	

The results show that the proposed framework achieves improvements across all metrics compared to the baseline strategy. Throughput (TP), defined as the total number of orders fulfilled during the simulation period, is increased by approximately 18.80%. This improvement highlights the benefit of near-future bin availability at each picking station and coordinating it with pod selection logic based on the collective order requirements at each picking station, thereby enabling higher order fulfillment capacity. Notably, isolated improvements are also observed when applying either the Future-Aware order assignment strategy or the Station-Match pod selection independently. The former yields an 5.48% increase in throughput, while the latter contributes a 6.84% improvement, underscoring the individual effectiveness of each decision module.

Regarding responsiveness, the proposed framework reduces the Order Completion Time (OCT) by 11.50% compared to the baseline configuration. This improvement reflects more efficient order processing, achieved through proactive order assignment that accounts for both current inventory levels and projected bin availability, enabling the system to identify picking stations that can fulfill orders more effectively. When evaluated independently, the proposed order assignment strategy results in a 5.41% reduction in OCT, while the enhanced pod selection strategy contributes a 2.28% improvement demonstrating that both modules contribute to increased responsiveness, even when applied individually.

In terms of pod utilization, the proposed model improves the pile-on rate by 34.86%, meaning that each pod carries and delivers a greater number of order lines per trip. This increase results from the application of the Station Match Strategy, which selects pods based on their ability to fulfill the combined SKU requirements of both active and pre-assigned orders at each picking station. When assessed independently, the proposed pod selection strategy yields a 32.22% improvement, while the Future-Aware POA contributes a 7.77% gain. These findings demonstrate that station SKU-based pod selection and Future-Aware order assignment independently contribute to optimizing pod utility.

The proposed framework enhances throughput capacity, reduces order completion time, and improves pod utilization. The proposed model has the following advantages:

- **Fulfillment-aware Pick Order Assignment (POA) with bin-level evaluation:**
The proposed Pick Order Assignment strategy improves decision quality by anticipating near-future bin availability at each picking station. This forward-looking capability enables the system to evaluate fulfillment potential across all picking stations, rather than relying on local or immediate conditions. As a result, orders are assigned to picking stations with higher fulfillment in a globally optimized manner, improving pod utilization and reducing overall order completion time.
- **Station-demand-oriented Pick Pod Selection (PPS):**
The Pick Pod Selection strategy evaluates candidate pods based on their ability to fulfilled the unfulfilled SKU demand of all active and pre-assigned orders at a given picking station. Instead of focusing on individual orders, the strategy considers the collective demand at the picking station to maximize pod utility by selecting pods that can contribute to multiple orders simultaneously.

4.2 Statistical Analysis

This study employed a comprehensive statistical approach to evaluate the performance differences between the baseline and proposed methods. The normality test was conducted following the hypotheses for the Shapiro–Wilk test. The null hypothesis assumes that the residuals are normally distributed, while the alternative hypothesis assumes they are not. The equality of variances was evaluated using Levene’s test, based on the standard hypothesis that the group variances are equal. The null hypothesis posits that the variances across all groups are equal, whereas the alternative hypothesis indicates that at least one group variance differs significantly.

4.2.1 Normality Test

Table 4.5 presents the results of the Shapiro–Wilk test for the three performances metrics, including throughput, order completion time, and pile-on, across the four evaluated scenarios. A p-value greater than 0.05 indicates that the data are normally distributed and do not violate the assumption of normality.

For the throughput metric, all scenarios including *PM_DM*, *PM_SM*, *FA_DM*, and *FA_SM* resulted in p-values greater than 0.05. This indicates that the throughput data can be considered normally distributed across all cases.

For the order completion time metric, the *FA_SM* scenario showed a p-value of 0.0002, which is less than 0.05, indicating a significant deviation from normality. Conversely, *PM_DM*, *PM_SM* and *FA_DM* scenarios yielded p-values above 0.05 and therefore met the assumption of normality.

For the pile-on metric, the *PM_DM* scenario failed the normality test with a p-values less than 0.05. The other scenarios, the *PM_SM*, *FA_DM*, and *FA_SM* scenarios passed the normality check.

Table 4.5 Shapiro–Wilk test results for normality

Metric	Model	W (Shapiro-Wilk)	p-value
Throughput	PM_DM	0.9634	> 0.05
Throughput	PM_SM	0.9734	> 0.05
Throughput	FA_DM	0.9733	> 0.05
Throughput	FA_SM	0.9741	> 0.05
Order Completion Time	PM_DM	0.3580	> 0.05
Order Completion Time	PM_SM	0.2341	> 0.05
Order Completion Time	FA_DM	0.2821	> 0.05
Order Completion Time	FA_SM	0.0002	< 0.05
Pile-on	PM_DM	0.8734	< 0.05
Pile-on	PM_SM	0.9333	> 0.05
Pile-on	FA_DM	0.9418	> 0.05
Pile-on	FA_SM	0.9435	> 0.05

The visual inspection of the Q–Q plots of Throughput, as shown in Figure 4.1, supports this observation. In each subplot, the sample quantiles closely follow the theoretical quantiles along the reference line, with only minor deviations at the tails. These patterns indicate no substantial skewness or departure from normality, thereby confirming that the throughput metric satisfies the normality assumption across all scenarios.

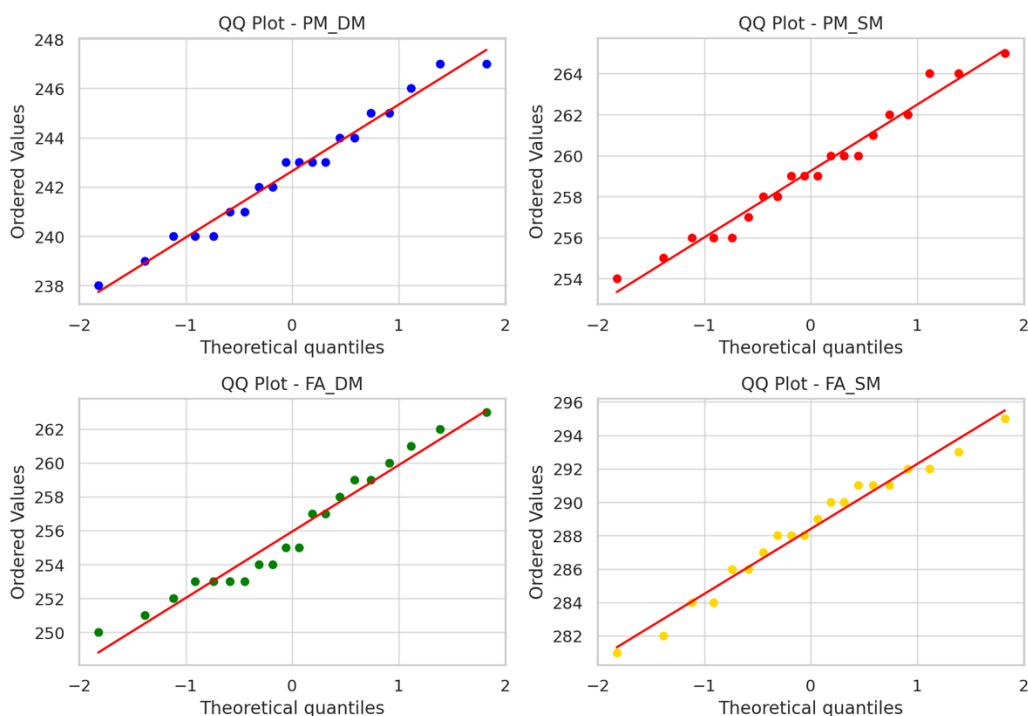


Figure 4.1 Q–Q plots of throughput for each scenario

4.2.2 Homogeneity of Variance Test

Levene's test was conducted to assess the assumption of homogeneity of variances for each metric. The results are summarized in Table 4.6. A p-value greater than 0.05 indicates that the variances among the groups are equal.

Table 4.6 Levene's test results for homogeneity of variances

Metric	Levene's F	p-value
Throughput	1.3605	> 0.05
Order Completion Time	0.8767	> 0.05
Pile-on	2.0446	> 0.05

The results of the assumption checks showed that the throughput metric fulfilled both the normality and equal variance criteria across all experimental groups. This confirms the suitability of applying a two-way ANOVA to analyze the effects of the tested strategies on throughput. In contrast, the order completion time and pile-on metrics violated the normality assumption in at least one scenario, despite satisfying the homogeneity of variances. To maintain the validity and robustness of the analysis, the Kruskal–Wallis test, a non-parametric alternative, was used for these two metrics.

4.2.3 Two-Way ANOVA Analysis of Throughput

A two-way Analysis of Variance (ANOVA) was conducted to examine the main and interaction effects of the Pick Order Assignment strategy (POA) and the Pick Pod Selection strategy (PPS) on the throughput metric. The results are presented in Table 4.7.

The main effect of POA was statistically significant, with an F-value of 807.45 and a p-value less than 0.001, indicating that the choice of order assignment strategy has a significant impact on throughput. Similarly, the main effect of PPS was also significant, with an F-value of 1078.05 and a p-value less than 0.001, suggesting that different pod selection strategies result in notable differences in throughput performance.

The interaction effect between POA and PPS was significant ($F = 112.57$, $p < 0.001$), indicating that the effect of one factor depends on the level of the other. These findings confirm that both individual strategies and their interaction have a significant influence on system throughput.

Table 4.7 Two-way ANOVA results for throughput

Factor	Sum Sq	df	F	p-value (PR(>F))
POA	9010.01	1	807.45	$p < 0.001$
PPS	12029.51	1	1078.05	$p < 0.001$
POA $\times$ PPS	1256.11	1	112.57	$p < 0.001$
Residual	848.05	76		

A Tukey Honestly Significant Difference (HSD) post hoc test was conducted to perform pairwise comparisons between the levels of the POA and PPS factors. The results are summarized in Table 4.8. For the POA factor, the comparison between the Future-Aware and Pod Match strategies revealed a statistically significant difference in mean throughput. The proposed strategy achieved a significantly higher mean throughput than the baseline, with a mean difference of 21.22 units ($p < 0.001$). The 95% confidence interval for this difference ranged from 15.23 to 27.21, indicating that the effect is both statistically and practically meaningful.

Table 4.8 Tukey HSD pairwise comparison results for throughput

Group 1	Group 2	Mean Diff	p-adj	95% CI Lower	95% CI Upper	Significant
Pod-Match	Future-Aware	21.22	0.0	15.23	27.21	Yes
Demand	Station Match	24.53	0.0	19.21	29.84	Yes

Similarly, for the PPS factor, the Station Match strategy resulted in significantly higher throughput compared to the Demand-based strategy. The observed mean difference was 24.53 units ($p < 0.001$), with a confidence interval ranging from 19.21 to

29.84. This finding further confirms that the Station Match strategy is more effective than the Demand-based approach in enhancing system throughput.

4.2.4 Non-Parametric Analysis of Order Completion Time

Due to violations of normality and homogeneity of variance assumptions, non-parametric statistical methods were applied to analyze the order completion time metric. A Kruskal–Wallis omnibus test was conducted to determine whether there were significant differences in order completion time across the four strategy combinations. The test result was $H\text{-statistic} = 68.98$ with a $p\text{-value}$ of 7.07×10^{-15} , indicating a statistically significant difference among the scenarios.

To evaluate the main effects of each proposed strategy, Mann–Whitney U tests were conducted to compare baseline and proposed model individually for POA and PPS. Results showed that both the Future-Aware POA and Station-Match PPS significantly reduced order completion time compared to their respective baselines, with $p\text{-value}$ less than 0.001, as summarized in Table 4.9.

Table 4.9 Main effects of POA and PPS on order completion time

Factor	Group 1	Group 2	U-statistic	p-value
POA main effect	Pod-Match	Future-Aware	345.0	< 0.001
PPS main effect	Demand	Station Match	400.0	< 0.001

Both comparisons yielded statistically significant results, indicating that the proposed Future-Aware order assignment and Station Match pod selection strategies contributed to reduced order completion time compared to their respective baselines.

Additionally, pairwise post-hoc comparisons were performed across all scenario combinations using the Mann–Whitney U test with Bonferroni correction. As shown in Table 4.10, all six comparisons remained statistically significant after adjustment ($p < 0.05$), shows of meaningful differences in order completion time among all strategy scenario.

Table 4.10 Post-Hoc Mann–Whitney U test results for order completion time

Group1	Group2	Bonferroni p	Significant
PM_DM	PM_SM	< 0.001	Yes
PM_DM	FA_DM	< 0.001	Yes
PM_DM	FA_SM	< 0.001	Yes
PM_SM	FA_DM	< 0.001	Yes
PM_SM	FA_SM	< 0.001	Yes
FA_DM	FA_SM	< 0.001	Yes

4.2.5 Non-Parametric Analysis of Pile-On

The pile-on metric was analyzed using non-parametric statistical methods due to a violation of the normality assumption. In particular, the Shapiro–Wilk test indicated that the *PM_DM* group did not follow a normal distribution. Accordingly, the Kruskal–Wallis H test was conducted to evaluate differences across the four experimental scenarios. The test yielded an H-statistic of 67.80 with a p-value of 1.26×10^{-14} , indicating a statistically significant difference in pile-on across the groups.

To assess the individual effects of the proposed strategies on pile-on performance, Mann–Whitney U tests were conducted to compare the baseline and proposed scenarios for both POA and PPS. As shown in Table 4.11, the proposed Future-Aware order assignment strategy significantly outperformed the baseline Pod Match strategy in terms of pile-on, with p-value less than 0.01. Similarly, the proposed Station Match pod selection strategy showed a highly significant improvement over the baseline Demand-based strategy, with p-value less than 0.001. These results confirm that both proposed strategies contribute meaningfully to increasing pod utilization efficiency.

Table 4.11 Main effects of POA and PPS on pile-on

Factor	Group 1	Group 2	U-statistic	p-value
POA main effect	Pod-Match	Future-Aware	512.0	< 0.01
PPS main effect	Demand	Station Match	0.0	< 0.001

To further examine specific group differences, pairwise comparisons were performed using the Mann–Whitney U test with Bonferroni correction. As summarized in Table 4.12, all pairwise comparisons yielded statistically significant results ($p < 0.05$ after adjustment), confirming that most scenario combinations led to significantly different pile-on performance.

Table 4.12 Post-Hoc Mann–Whitney U test results for pile-on

Group1	Group2	Bonferroni p	Significant
PM_DM	PM_SM	< 0.001	Yes
PM_DM	FA_DM	< 0.001	Yes
PM_DM	FA_SM	< 0.001	Yes
PM_SM	FA_DM	< 0.001	Yes
PM_SM	FA_SM	< 0.05	Yes
FA_DM	FA_SM	< 0.001	Yes

4.3 Sensitivity Analysis

In this study, it is recognized that variations in key parameter values can significantly influence overall system performance. Therefore, a sensitivity analysis is conducted to evaluate how changes in input variables impact the resulting system outcomes. The analysis focuses on three critical parameters: the number of robots (AGVs), the number of pods, and the number of picking stations. For each parameter, its value is adjusted independently while the other parameters are held constant at the reference values specified by the system configuration. The detailed numerical results of this sensitivity analysis are presented in Table 4.13.

Table 4.13 Sensitivity analysis results

Parameters	Value	Performance		
		Throughput	Order Completion Time (Second)	Pile-on
Number of Robots	10	256	399.23	7.5
	15	273	334.67	8.1
	20	289	311.64	8.5
	25	295	332.42	8.4
	30	281	344.29	8.3
	35	275	358.59	8.0
	40	253	382.61	7.7
Number of Pods	380	272	303.57	8.0
	400	281	310.39	8.3
	420	288	308.42	8.4
	440	284	344.71	8.3
	460	269	357.09	8.1
Number of Picking Stations	2	256	275.57	8.1
	3	289	299.79	8.4
	4	296	342.13	8.2
	5	293	368.92	8.2

4.3.1 The Effect of the Total Number of Robots

The total number of robots deployed in the warehouse significantly influences system performance. As shown in Figure 4.2, increasing the number of robots from 10 to 20 results in a notable improvement in throughput, rising from 256 to 289, an approximate 13% increase. Simultaneously, the order completion time decreases from 399.23 to 311.64 seconds, representing a 22% improvement, while the pile-on increases from 7.5 to 8.5. This improvement is attributed to improved parallelization of pod deliveries, which reduces picker idle time and accelerates order fulfillment.

However, beyond 20 robots, the performance gains begin to diminish. Increasing the robot count to 25 yields only a 2% increase in throughput, reaching 295, while order completion time slightly increases to 332.42 seconds and pile-on decreases slightly to 8.4. As the number of robots rises to 30 and 35, throughput declines to 281 and 275 respectively, order completion time increases to 344.29 and 358.59 seconds, and pile-on drops to 8.3 and 8.0. At 40 robots, the trend worsens, with throughput declining to 253, order completion time rising to 382.61 seconds, and pile-on decreasing further to 7.7.

This finding indicates that deploying too many robots can lead to traffic congestion in warehouse aisles, causing bottlenecks as multiple robot queue for limited picking stations. This congestion reduces routing flexibility, increases waiting time, and lowers overall throughput efficiency. In addition, an oversized fleet elevates operational costs, including capital investment, energy consumption, and maintenance. Therefore, while moderate robot deployment enhances system performance, excessive robot counts introduce inefficiencies. The number of robots should be carefully optimized to balance throughput improvement with congestion risks and cost-effectiveness.

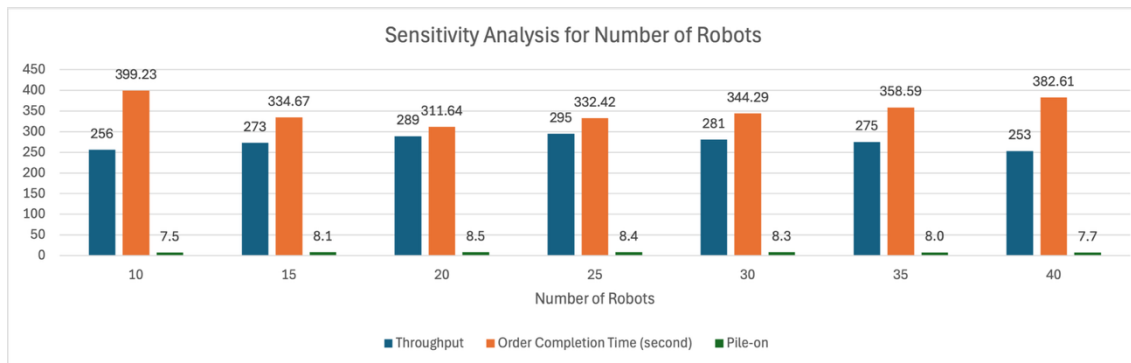


Figure 4.2 Impact of number of robots on system performance metrics

4.3.2 The Effect of the Total Number of Pods

The total number of pods in the system determines SKU availability and impacts the system's flexibility and robot routing efficiency. A sufficient number of pods ensures sufficient SKU across pods, providing multiple options for fulfilling orders.

As shown in Figure 4.3, increasing the number of pods from 380 to 420 leads to a 5.9 % improvement in throughput, rising from 272 to 288, and a modest reduction in order completion time from 303.57 to 308.42 seconds. Pile-on also improves slightly from 8.0 to 8.4, reflecting increased SKU redundancy and better pod to station demand matching. These improvements are attributed to more flexible fulfillment opportunities and higher SKU availability across pods, enabling more efficient pod utilization and reduced fulfillment delays. However, beyond 420 pods, marginal gains begin to diminish. At 440 pods, throughput decreases slightly to 284, while order completion time increases significantly to 344.71 seconds, despite pile-on remaining stable at 8.3. With 460 pods, the performance continues to decline: throughput drops further to 269, order completion time increases to 357.09 seconds, and pile-on falls to 8.1.

Operating with too many pods are stored in the available space, robot navigation becomes constrained due to limited movement space. The congestion within the storage zones can force robots to take longer routes, increasing travel time and potentially causing delays in pod returns. In contrast, when the number of pods is decrease, robots benefit from increased free space and greater routing flexibility. However, limited SKU redundancy reduces the system's ability to match orders efficiently. Therefore, the number of pods should be determined to maintain an effective balance between SKU availability and storage capacity within the warehouse.

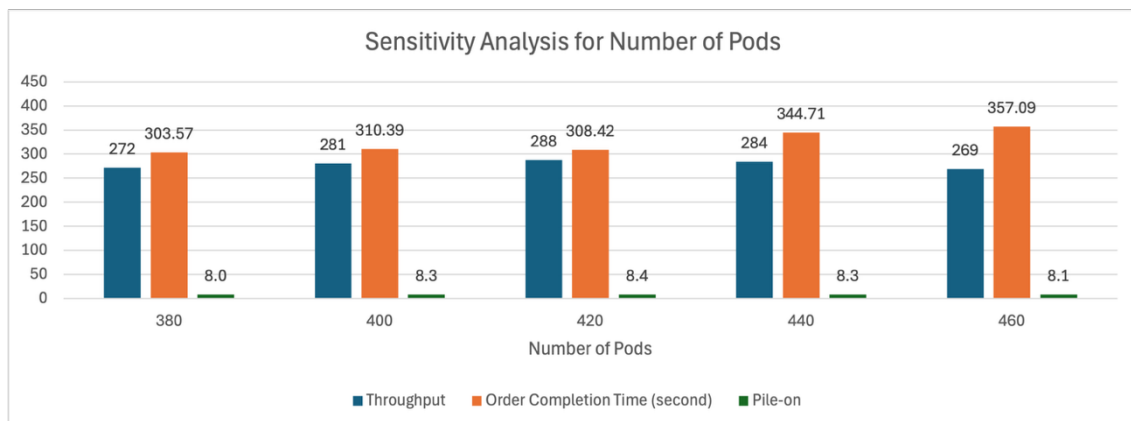


Figure 4.3 Impact of number of pods on system performance metrics

4.3.3 Effect of the Number of Picking Stations

The sensitivity analysis results show that increasing the number of picking stations generally improves system performance by allowing more orders to be process simultaneously. As shown in Figure 4.4, increasing the number of picking stations from 2 to 3 leads to a notable improvement in throughput, rising from 256 to 289, representing a 13% increase. Order completion time also increases moderately from 275.57 to 299.79 seconds, while pile-on improves from 8.1 to 8.4. Further increasing the number of picking stations from 3 to 4 results in a smaller throughput gain of 2 %, reaching 296, with order completion time increasing to 342.13 seconds and pile-on slightly declining to 8.2. Further increasing the number of picking stations from 4 to 5, throughput slightly declines to 293, while order completion time further rises to 368.92 seconds. This indicates a decline in efficiency caused by adding more picking stations without a corresponding increase in the number of robots can lead to unintended inefficiencies. Pickers may experience increased idle time as they wait for the necessary pods to arrive, resulting in underutilized picking stations.

However, when the number of picking stations increases from 4 to 5, throughput slightly declines to 293, while order completion time further rises to 368.92 seconds. This indicates a decline in efficiency caused by adding more picking stations without a corresponding increase in the number of robots can lead to unintended inefficiencies. Pickers may experience increased idle time as they wait for the necessary pods to arrive, resulting in underutilized picking stations.

These findings suggest that the number of picking stations should be scaled in coordination with robot fleet size, pod availability, order volume, and warehouse size to ensure balanced and efficient system performance. A well-balanced relationship between robots and picking stations is essential to maintaining high system performance and minimizing labor idle time.

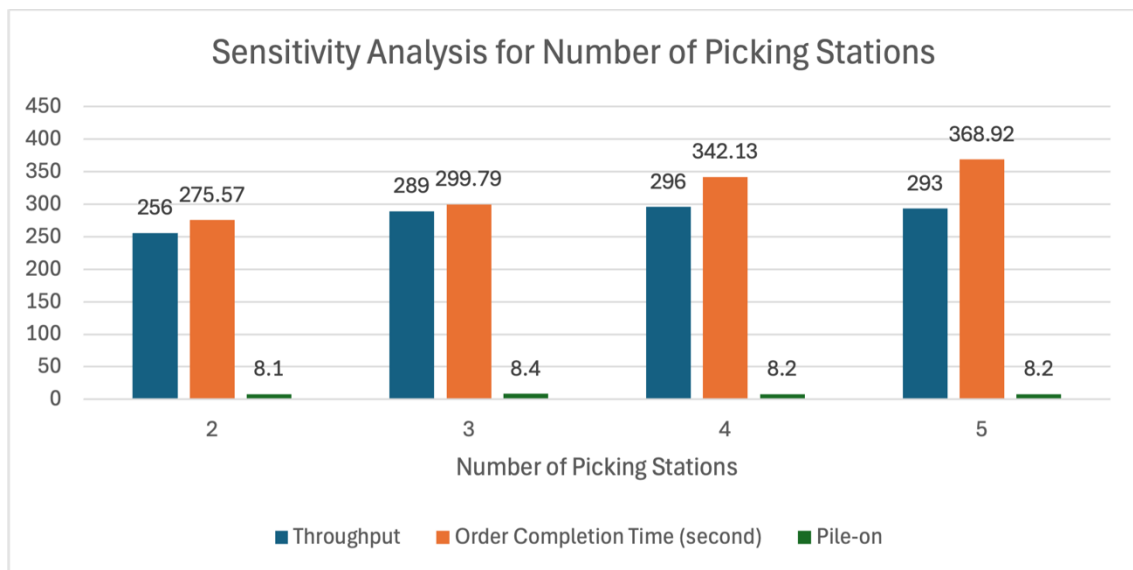


Figure 4.4 Impact of number of picking station on system performance metrics

CHAPTER 5

CONCLUSION & FUTURE WORK

5.1 Conclusion

This study focused on the operational stages of Pick Order Assignment (POA) and Pick Pod Selection (PPS) in a Robotic Mobile Fulfillment System (RMFS) operating under real-time constraints and multiple SKU conditions. This study presents several key contributions. First, the proposed Future-Aware POA dynamically assign orders by evaluating both current and projected bin availability, and projected SKU fulfillment based on the inventory of active pods. This approach enabling more informed decisions that enhances picking efficiency by achieving higher throughput and increases the overall responsiveness of system. For pod selection, the station-match strategy aggregates SKU demand across all active orders at each picking station, adopting a demand-driven approach that selects pods based on their ability to fulfill the station's remaining unit demand. The approach leading to improves operational effectiveness by increasing the number of order lines fulfilled per pod visit. Additionally, the study implements a nearest-location return policy for robots within the NetLogo simulation environment. This return mechanism captures the dynamic nature of warehouse operations by continuously adapting to evolving storage conditions and spatial configurations.

The simulation model, developed using NetLogo, captured essential aspects of RMFS dynamics, including real-time bin availability, robot activity, and inventory levels. The simulation was executed over an eight-hour and replicated 20 times to ensure robustness of the results. Experimental findings indicate that the proposed strategies consistently outperform baseline configurations, achieve higher throughput, reduce order completion times, and improve pile-on.

5.2 Future Research

This study demonstrates that the staged, Future-Aware strategy can significantly improve RMFS performance. Nevertheless, several opportunities remain for extending and refining the current approach. One potential direction is to extend the PPS strategy to incorporate multi-robot and multi-drop-off configurations to enable many-to-many interaction among robots, pods, and picking stations. This enhancement facilitates more advanced coordination mechanisms between pod retrieval and return, allowing pods to dynamically matched with robots and routed to appropriate picking stations. These developments are expected to improve task distribution efficiency and reinforce the system's scalability and responsiveness under dynamic operational conditions.

Another direction involves incorporating SKU turnover rates and future order demand into the decision-making process. By prioritizing pods that contain high-turnover SKUs likely to be picked again in the near future, the system can reposition those pods closer to picking stations.

Incorporating pod face orientation into the pod selection process is an important consideration in warehouse operations. As each pod may have multiple accessible sides, it becomes necessary to determine which face should be directed toward the picking station to improve access to the required SKUs. Selecting the appropriate face can enhance picking efficiency by minimizing handling time and increasing SKU visibility.

Additionally, including robot battery management and charging scheduling would make the framework more realistic and applicable to industrial settings. Energy-aware routing and task scheduling allows the system to balance robot workloads effectively while ensuring timely access to charging stations. This approach helps maintain continuous robot availability and minimizes the risk of unexpected downtime, particularly during periods of high operational demand.

Finally, evaluating the proposed model under larger and more complex simulations would help confirm the scalability and generalizability of the staged approach. Adding the ability to handle urgent or priority orders would make the system more responsive to dynamic and time-sensitive operational demands. These directions present valuable opportunities for future research to enhance the practicality and adaptability of robotic mobile fulfillment systems in real-world warehouse environments.

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